Data Visualization Associate Early Internship

# SLU 0408 DVA Team 40

1st Week Final Exploratory Data Analysis (EDA) Report

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## Week 1 – Exploratory Data Analysis (Team Report)

**Internship:** Data Visualization Associate (Excelerate Virtual Internship)
**Week: 1** – Data Familiarization, PostgreSQL Setup & Exploratory Data Analysis
**Team Contribution:** Joint report covering analysis of all six assigned datasets

### Introduction

During the first week of the Excelerate Data Visualization Internship, our team focused on familiarizing ourselves with six distinct datasets supplied by the program. This initial phase was crucial for gaining a clear understanding of the data, and preparing it for further processing and visualization in the following weeks.

The objectives for this week included:

- Set up PostgreSQL and load the datasets
- Familiarize ourselves with the data structure, fields, types, and relationships
- Perform Exploratory Data Analysis (EDA) using SQL queries
- Summarize trends, inconsistencies, and key statistics
- Document insights in a structured EDA report

### List of Assigned Datasets

1. User_data.csv
2. Opportunity_Raw.csv
3. Cohort_Raw.csv
4. Marketing_data.csv
5. Learner_Opportunity_Raw.xlsx
6. Cognito_Raw2.xlsx

Each dataset was designated to a team member for thorough analysis and reporting. The upcoming sections will showcase individual contributions, exploratory data analysis (EDA) findings, along with relevant screenshots and visualizations.

# 1st Week Final EDA Report

Data Visualization Associate Early Internship

**Dataset: User Data (user_data.csv)**

**Phase1:DATA FAMILIARIZATION**

## Understanding Data Sources:

The dataset contains learner profile data, including columns for

**learner_id**, **country**, **degree**, **institution**, and **major**. This type of information is typically collected during a registration process, suggesting the data was likely exported from a learner registration system, a student information system, or a CRM (Customer Relationship Management) database.

A significant amount of data is missing, especially in the **degree**, **institution**, and **major fields**. These fields also use free-form text, which can lead to inconsistencies and potential biases.

| learner_id | country | degree | institution | major |
|---|---|---|---|---|
| Learner#00004f18 | Nigeria | Undergraduate Student | Federal University of Techn | Civil Engineering |
| Learner#0000647 | Nigeria | NULL | NULL | NULL |
| Learner#0001056 | Kenya | Graduate Student | UNICAF UNIVERSITY | Environmental Sustainability |
| Learner#00011c8 | Bangladesh | NULL | NULL | NULL |
| Learner#000141a | Nigeria | NULL | NULL | NULL |
| Learner#0acf3501 | Ghana | NULL | NULL | NULL |
| Learner#0001ca2 | Nigeria | Graduate Student | Nasarawa State University, | Accounting |
| Learner#0003bed | Pakistan | Graduate Student | CTTI College KP Campus | Part Time |
| Learner#0004295 | Nigeria | Undergraduate Student | Caleb University | Computer Science |
| Learner#00049a8 | Philippines | Undergraduate Student | Our lady of fatima universit | Nursing |
| Learner#0005243 | Pakistan | Graduate Student | PGMI | Doctor |
| Learner#00064ab | India | NULL | NULL | NULL |
| Learner#000687a | India | NULL | NULL | NULL |
| Learner#000693c | India | NULL | NULL | NULL |
| Learner#00070ea | Nigeria | NULL | NULL | NULL |
| Learner#00075a4 | Nigeria | Graduate Student | Durham University | Finance |
| Learner#0008438 | India | High School Student | Saket International School | Medicine |
| Learner#000893c | United Arab E | NULL | NULL | NULL |
| Learner#000926e | Pakistan | Graduate Student | Air University | Marketing |
| Learner#0009b4f7 | India | NULL | NULL | NULL |
| Learner#0009e8a | Nepal | Undergraduate Student | Global college internationa | Travel and Tourism |
| Learner#000aade | Pakistan | NULL | NULL | NULL |
| Learner#000b6d8 | India | NULL | NULL | NULL |
| Learner#000bce0 | India | Undergraduate Student | MG University | Cyber Security |
| Learner#000e027 | Pakistan | Undergraduate Student | FAST NUCES, Islamabad | Computer Science |

## Review Column Meanings and Data Types:

| Column Name | Data Type | Observations |
|---|---|---|
| learner_id | Text/Identifier | A unique identifier for each learner. The data type is a string, which is appropriate. |
| country | Character Varying | Represents the learner's country. The data type is a string. Some values are missing. |
| degree | Character Varying | Describes the learner's degree level. The data type is a string. A significant number of values are missing. |
| institution | Character Varying | The educational institution the learner is associated with. The data type is a string. A significant number of values are missing. |
| major | Character Varying | The learner's field of study or major. The data type is a string. A significant number of values are missing. |

## Identify Key Fields and Relationships:

**Primary Key:**

The **primary key** in this dataset is the learner_id column. This field serves as a unique identifier for each individual learner, ensuring that every record can be distinctly identified. A check of the data confirms that all 129,259 values in this column are unique.

**Foreign Keys:**

Based on the single file provided (Learner_Raw.csv), there are **no foreign keys**. Foreign keys are used to link data across multiple tables.

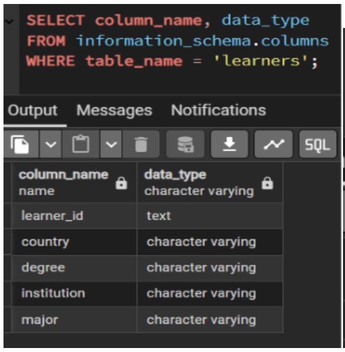
**Table Relationships:**

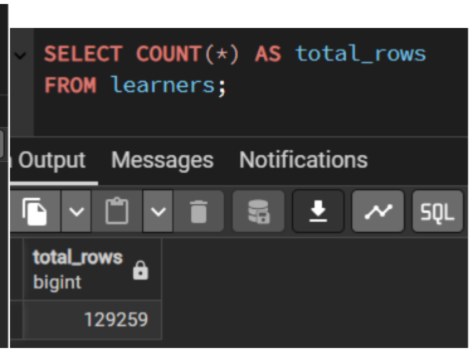
Since there is only one table, there are no relationships to describe.

## Perform an Initial Data Scan:

The Learner_Raw.csv file has the following characteristics:

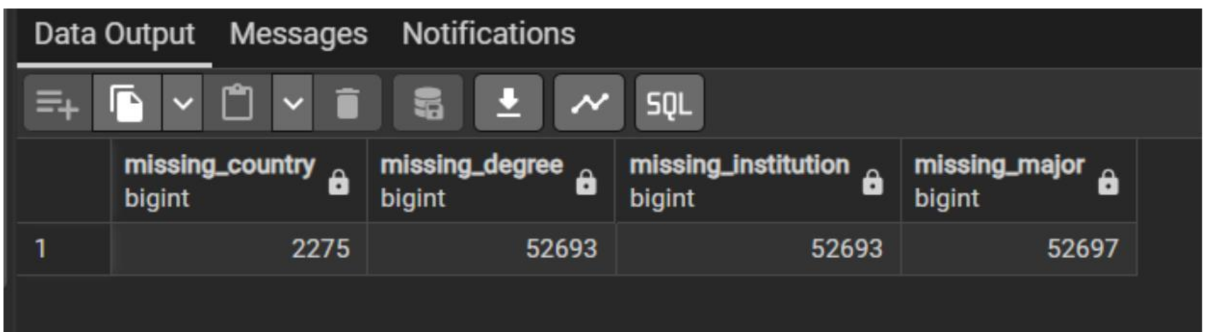
**Rows and Columns:** The dataset has 129,259 rows and 5 columns.




**Missing Values**: There are significant missing values in several columns:

- `country`: 2,275 missing values.
- `degree`: 52,693 missing values.
- `institution`: 52,693 missing values.
- `major`: 52,697 missing values.




## Summary of Findings:

**Key Observations:**

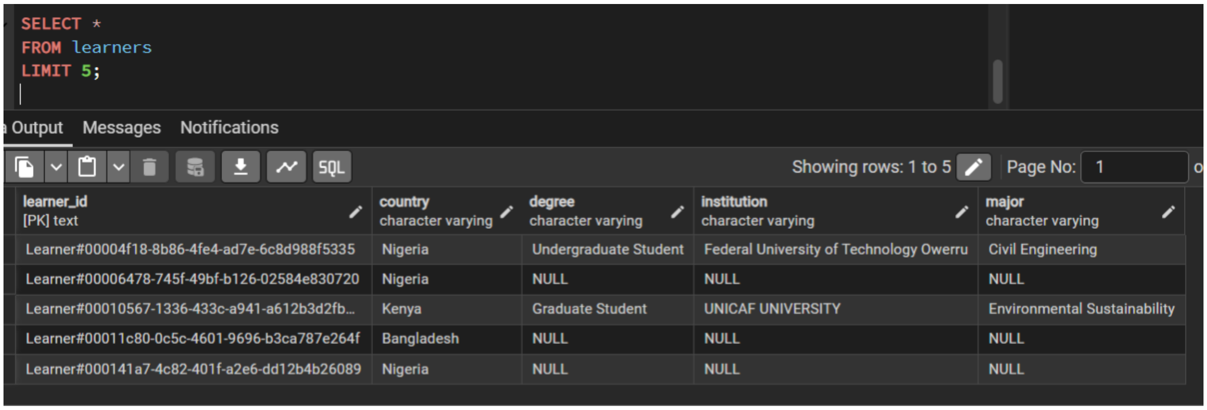
- **Data Source:** The Learner_Raw.csv file appears to be a learner profile data, likely exported from a CRM or a similar learner management system.
- **Data Volume:** The dataset contains **129,259 rows** and **5 columns**.
- **Data Quality Issues:**
  - o **Missing Values:** There's a high number of missing values, particularly in the degree, institution, and major columns (over 52,000 in each).

- **Duplicates:** The dataset contains a significant number of duplicate rows (**52,692**). This is a major issue that needs to be addressed before any analysis can be performed.

## Phase 2: Exploratory Data Analysis (EDA)

### Step 1: Understanding the Dataset:

1. **View the first few rows:**



```
SELECT *
FROM learners
LIMIT 5;
```

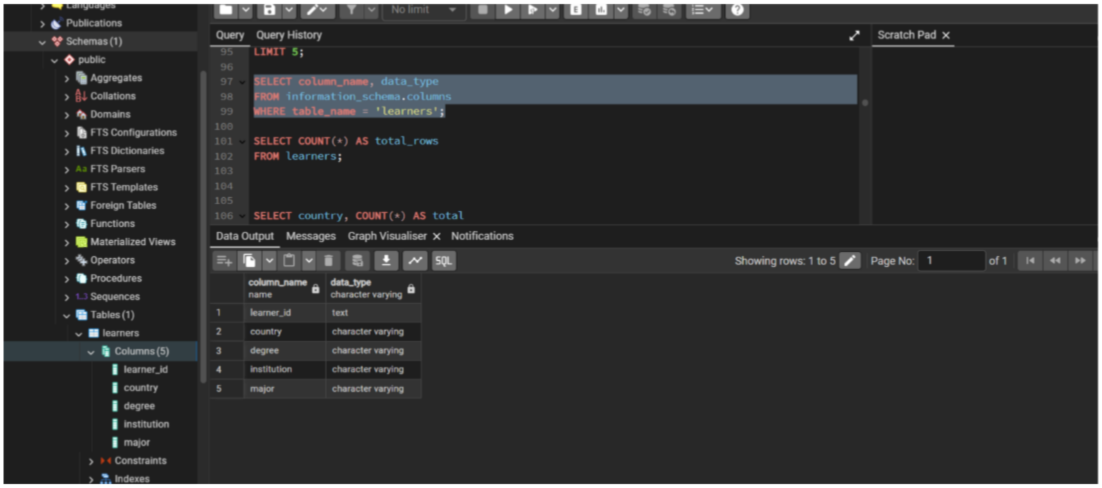
Output Messages Notifications

Showing rows: 1 to 5 Page No: 1

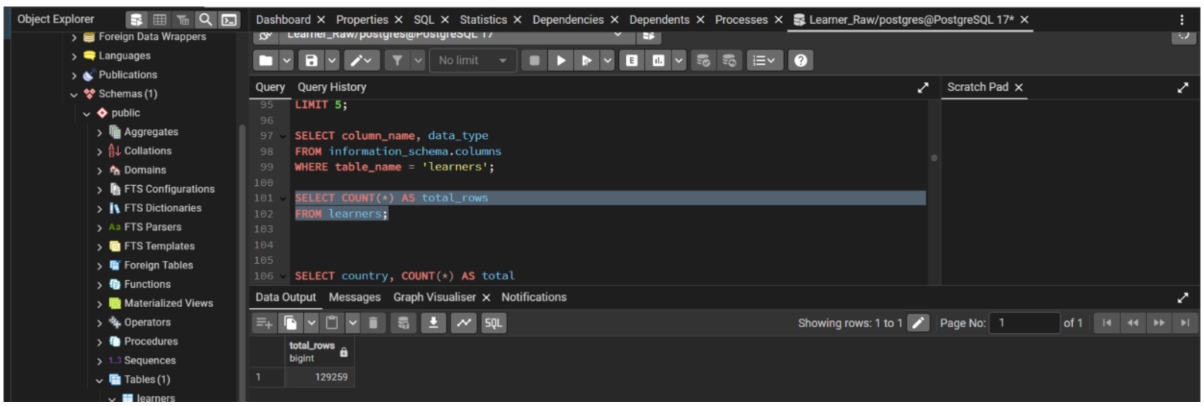
| learner_id [PK] text | country character varying | degree character varying | institution character varying | major character varying |
|---|---|---|---|---|
| Learner#00004f18-8b86-4fe4-ad7e-6c8d988f5335 | Nigeria | Undergraduate Student | Federal University of Technology Owerru | Civil Engineering |
| Learner#00006478-745f-49bf-b126-02584e830720 | Nigeria | NULL | NULL | NULL |
| Learner#00010567-1336-433c-a941-a612b3d2fb... | Kenya | Graduate Student | UNICAF UNIVERSITY | Environmental Sustainability |
| Learner#00011c80-0c5c-4601-9696-b3ca787e264f | Bangladesh | NULL | NULL | NULL |
| Learner#000141a7-4c82-401f-a2e6-dd12b4b26089 | Nigeria | NULL | NULL | NULL |

2. **Check column names, data types, and structure:**

- There are 5 columns i.e. ("**learner_id**"(PK), "**country**", "**degree**", "**institution**", "**major**"
- The data types of the Columns are:
  - **learner_id =** Text
  - **country =** Character varying
  - **degree =** Character varying
  - **institution =** Character varying
  - **major =** Character varying




3. **Count total number of records:**
    - There are 129259 rows in the dataset




## <u>Step 2: Summarize the Data:</u>

The columns that are in the Dataset (learner_id, country, degree, institution, major), can't calculate:

- **Mean** (average)
- **Median**
- **Min / Max** (for numbers)
- **Standard deviation**

because all the fields are **text or IDs**, not numeric measures.

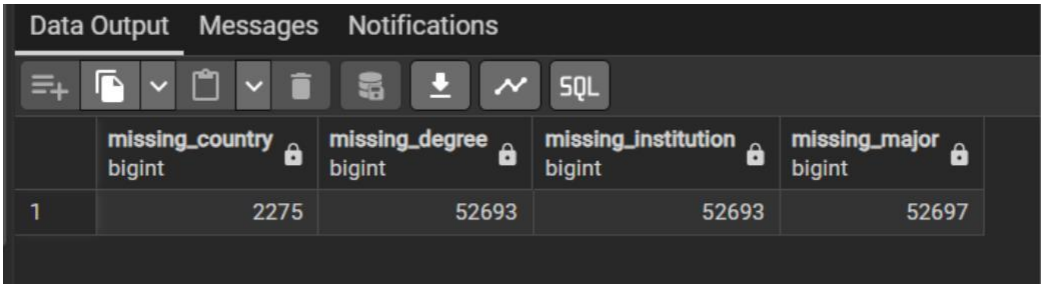
Those statistics only make sense for numeric data for example:

- Exam scores
- Ages
- Salaries
- Years of experience

## Step 3: Identify Missing and Duplicate Values:

**Missing Values**: There are significant missing values in several columns:

- `country`: 2,275 missing values.
- `degree`: 52,693 missing values.
- `institution`: 52,693 missing values.
- `major`: 52,697 missing values.



| | missing_country bigint | missing_degree bigint | missing_institution bigint | missing_major bigint |
|---|---|---|---|---|
| 1 | 2275 | 52693 | 52693 | 52697 |

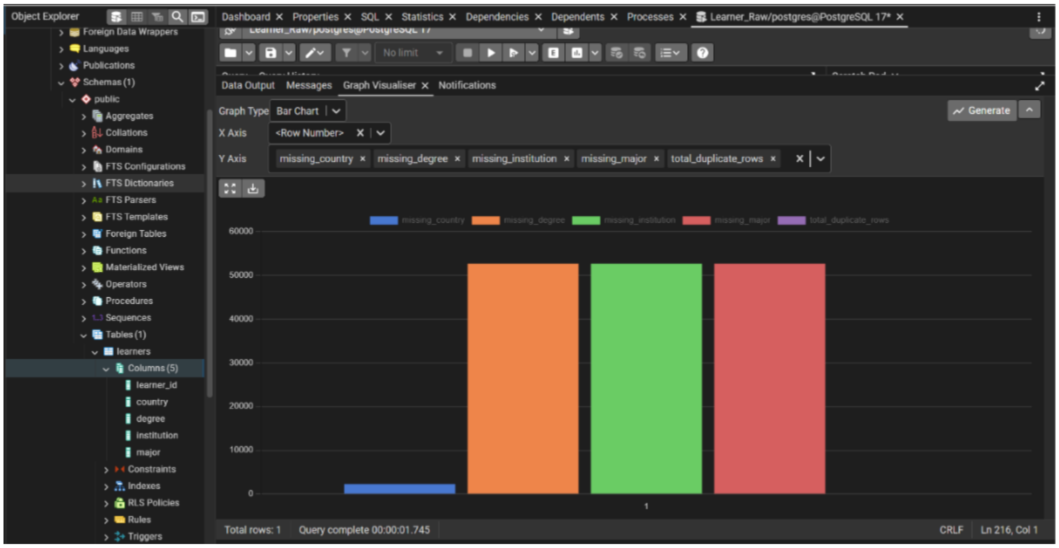
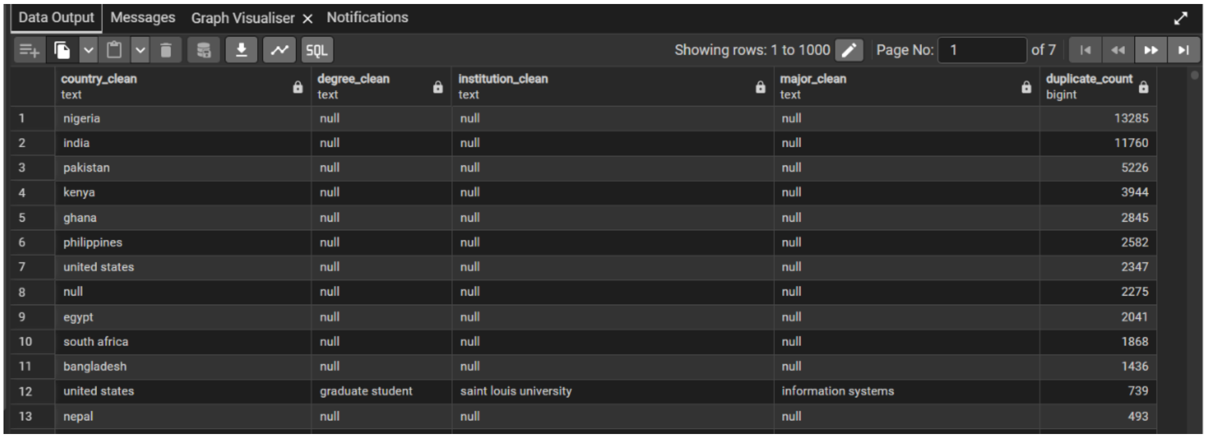



Figure: Bar Chart Of “Missing” Values in each Column

# Detecting Duplicate Records:

Possible duplicates by treating **country**, **degree**, **institution**, and **major** as the identifying combination



| | country_clean text | degree_clean text | institution_clean text | major_clean text | duplicate_count bigint |
|---|---|---|---|---|---|
| 1 | nigeria | null | null | null | 13285 |
| 2 | india | null | null | null | 11760 |
| 3 | pakistan | null | null | null | 5226 |
| 4 | kenya | null | null | null | 3944 |
| 5 | ghana | null | null | null | 2845 |
| 6 | philippines | null | null | null | 2582 |
| 7 | united states | null | null | null | 2347 |
| 8 | null | null | null | null | 2275 |
| 9 | egypt | null | null | null | 2041 |
| 10 | south africa | null | null | null | 1868 |
| 11 | bangladesh | null | null | null | 1436 |
| 12 | united states | graduate student | saint louis university | information systems | 739 |
| 13 | nepal | null | null | null | 493 |

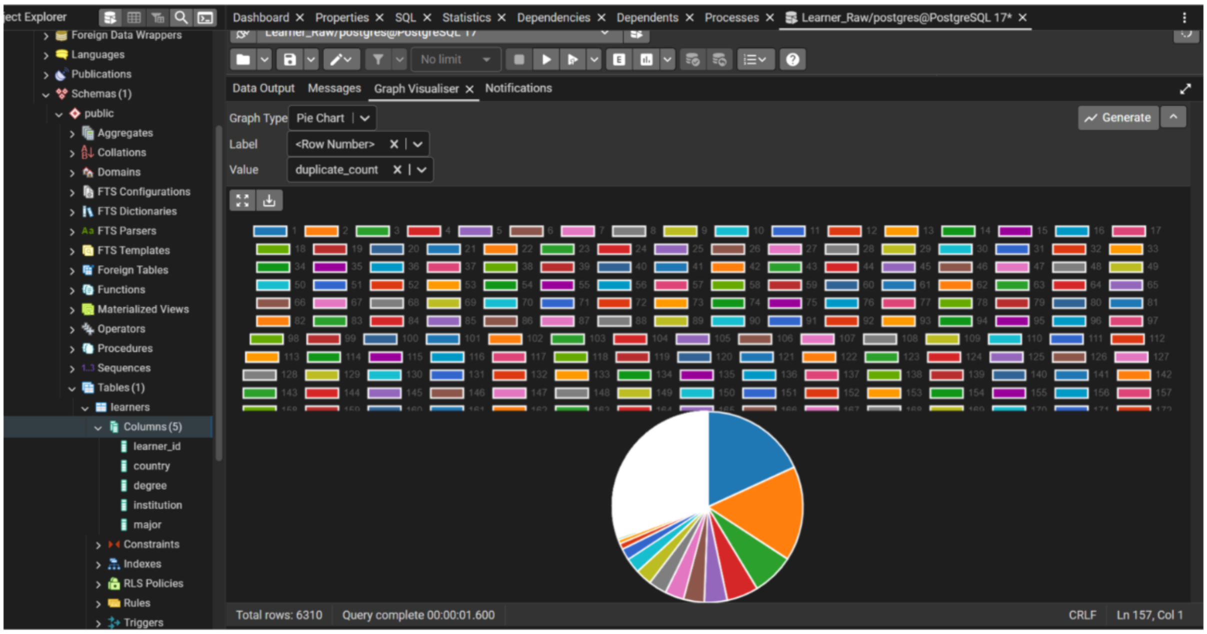
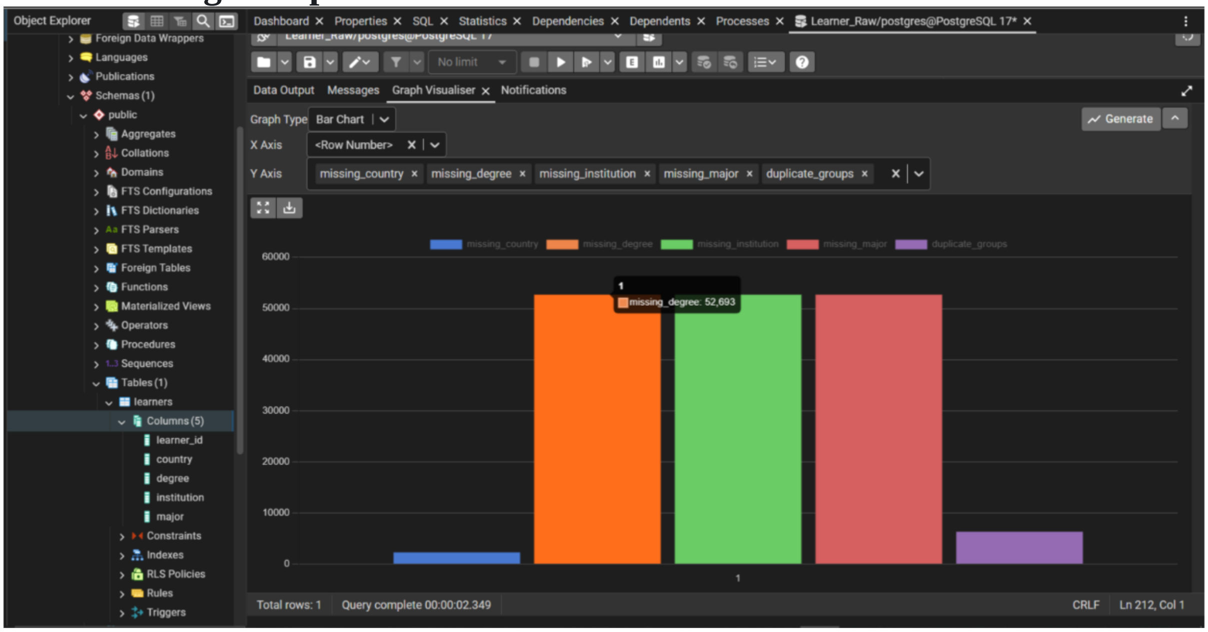



Figure: Pie Chart for Detecting Duplicate Records

## Total Missing + Duplicates:

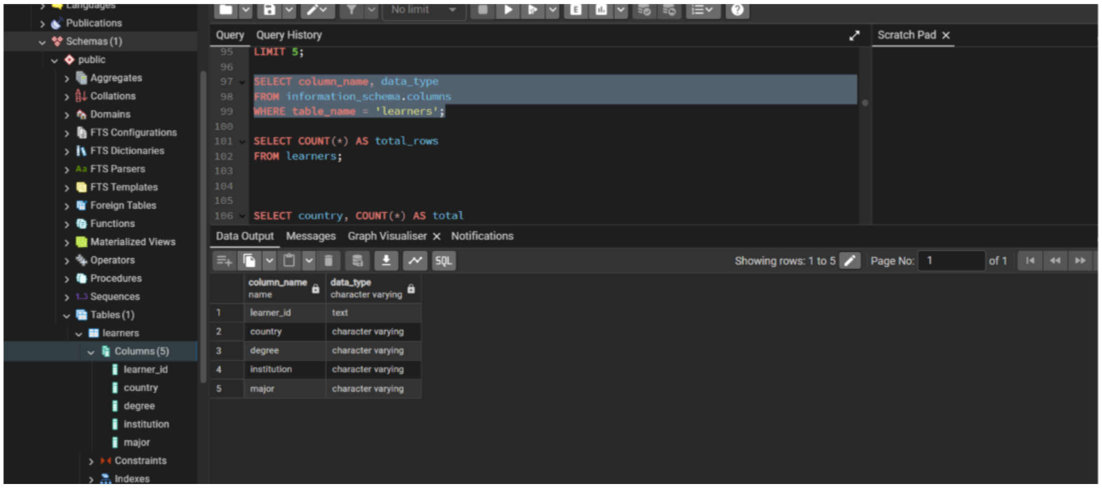



| | missing_country bigint | missing_degree bigint | missing_institution bigint | missing_major bigint | duplicate_groups bigint |
|---|---|---|---|---|---|
| 1 | 2275 | 52693 | 52693 | 52697 | 6310 |

Figure: Bar Chart for Total Missing + Duplicates

## Step 4: Spot Outliers and Anomalies:

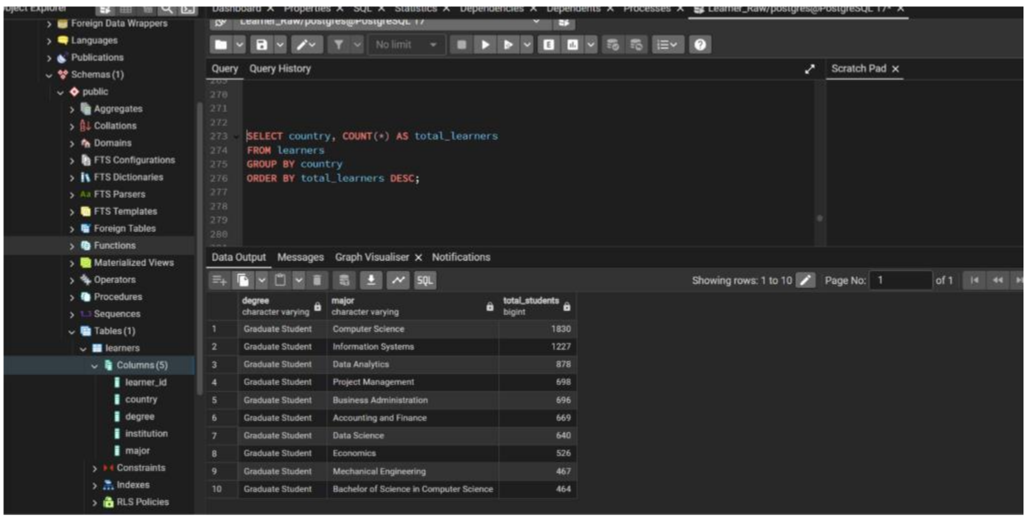
Table doesn’t have a numeric column, the IQR method only works on numeric columns (e.g., age, GPA, salary, marks, etc.).




## Step 5: Visualize Data Trends:

### Histogram Graph:

This bar chart displays the top 10 countries represented in the dataset, showing the number of learners from each country. The x-axis lists countries, while the y-axis shows the total learner count.



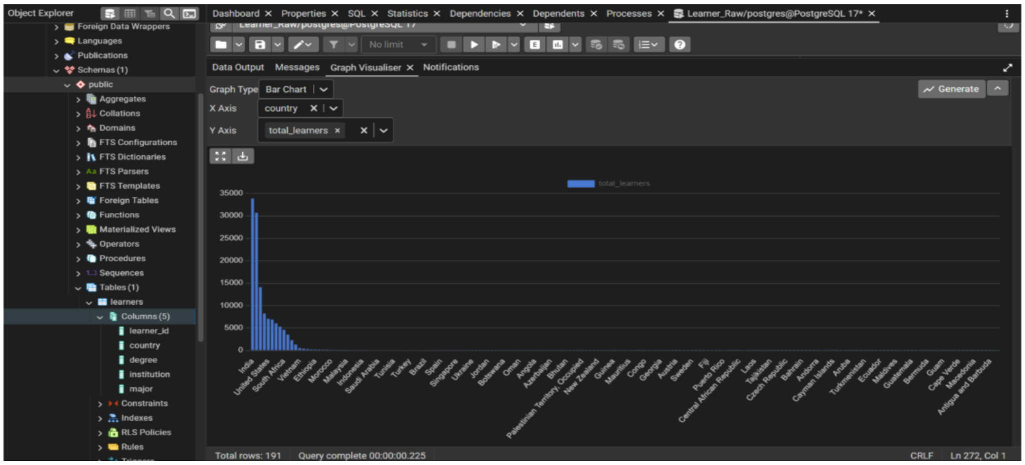



Figure: Bar Chart for Top 10 Countries by Number of Learners

## Line charts Graph:

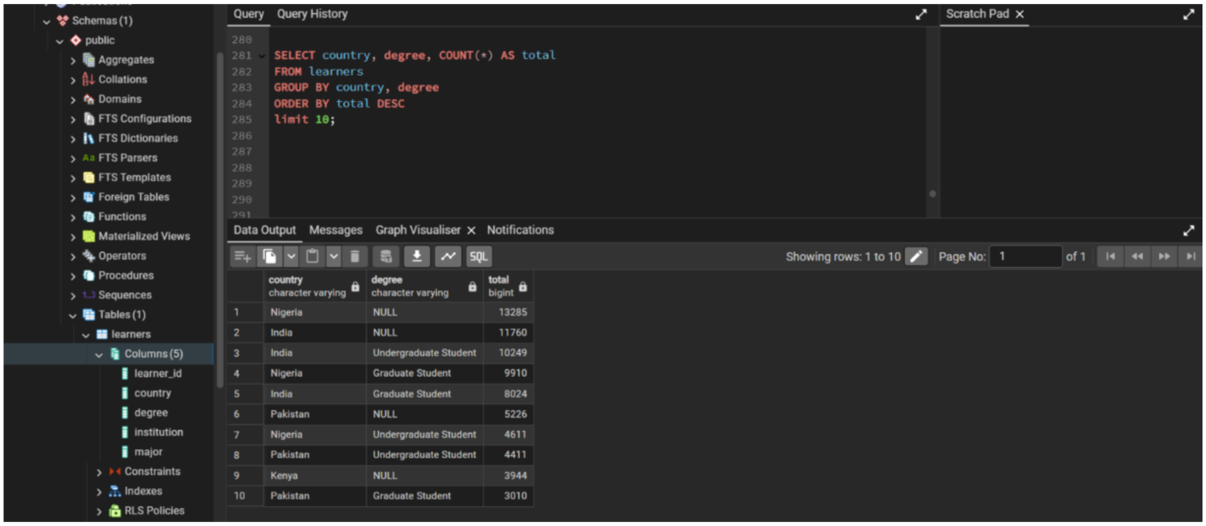
The dataset doesn't have any numeric columns, a traditional line chart won't work, because line charts need continuous numerical values to plot trends over time or an ordered sequence.

## Box plots Graph:

A box plot was generated to visualize the distribution and identify outliers in the dataset's numeric columns. The dataset doesn't have any numeric columns.

## Relationship Analysis – Compare variables

- Learners by country & degree:



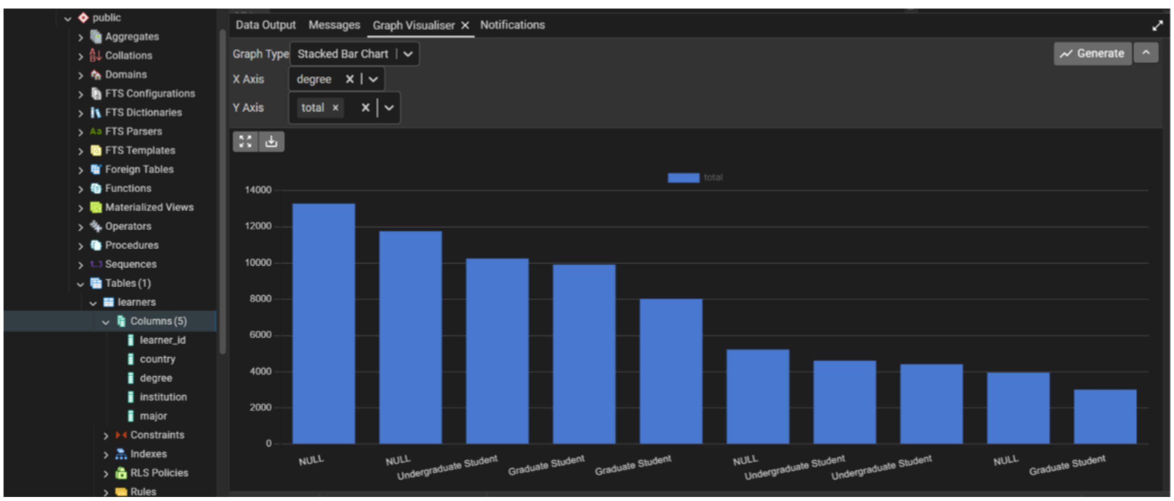
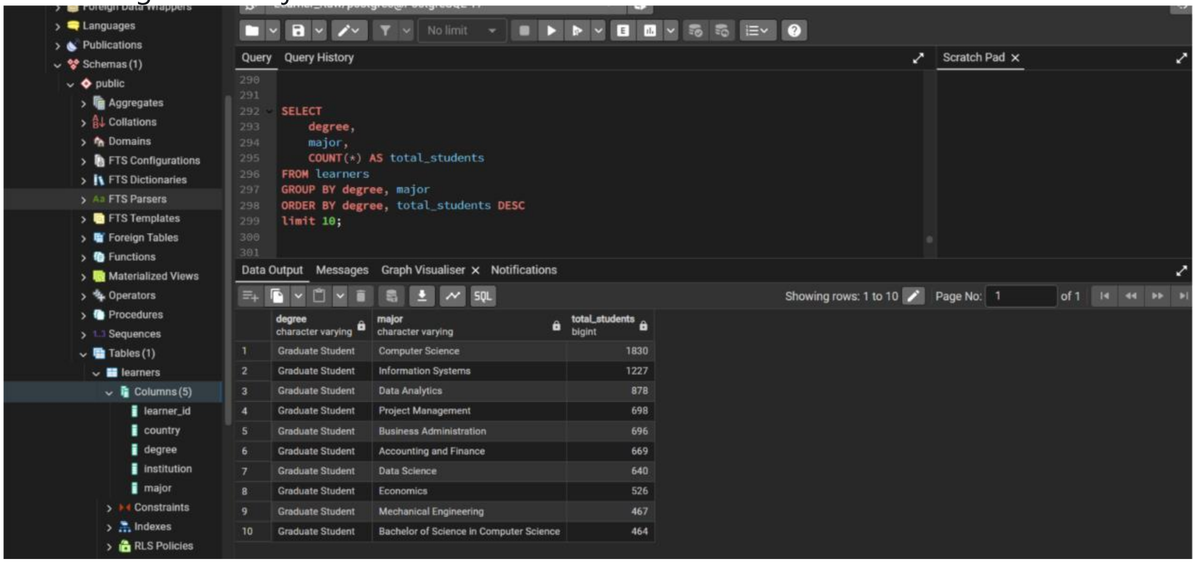




Figure: Stacked Bar Chart for Distribution of Learners by Country and Degree

- Degree VS Major:



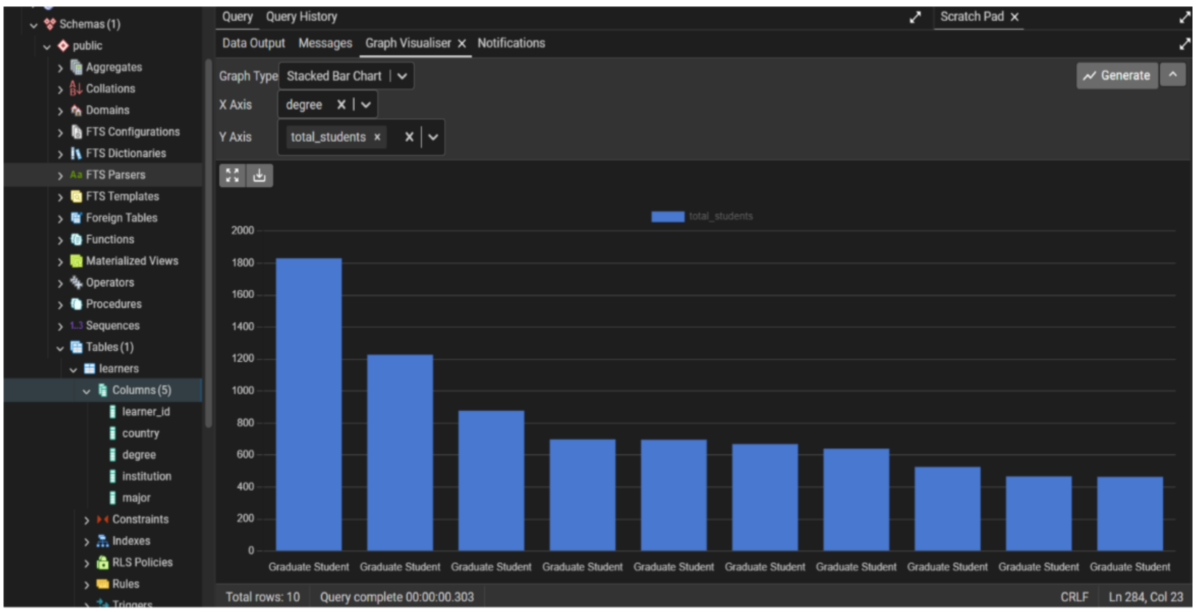
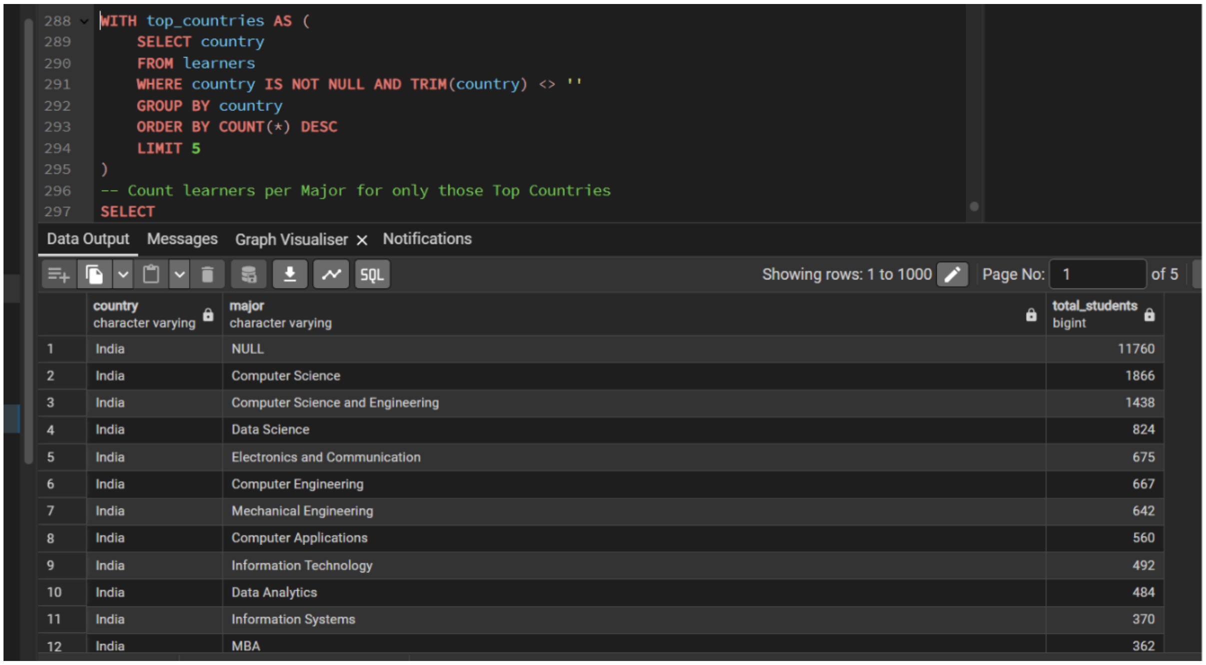




Figure: Stacked Bar Chart for Comparison of Degree vs. Major Among Learners

- Top 5 Countries-Major Distribution:



```
WITH top_countries AS (
    SELECT country
    FROM learners
    WHERE country IS NOT NULL AND TRIM(country) <> ''
    GROUP BY country
    ORDER BY COUNT(*) DESC
    LIMIT 5
)
-- Count learners per Major for only those Top Countries
SELECT
```

| | country character varying | major character varying | total_students bigint |
|---|---|---|---|
| 1 | India | NULL | 11760 |
| 2 | India | Computer Science | 1866 |
| 3 | India | Computer Science and Engineering | 1438 |
| 4 | India | Data Science | 824 |
| 5 | India | Electronics and Communication | 675 |
| 6 | India | Computer Engineering | 667 |
| 7 | India | Mechanical Engineering | 642 |
| 8 | India | Computer Applications | 560 |
| 9 | India | Information Technology | 492 |
| 10 | India | Data Analytics | 484 |
| 11 | India | Information Systems | 370 |
| 12 | India | MBA | 362 |

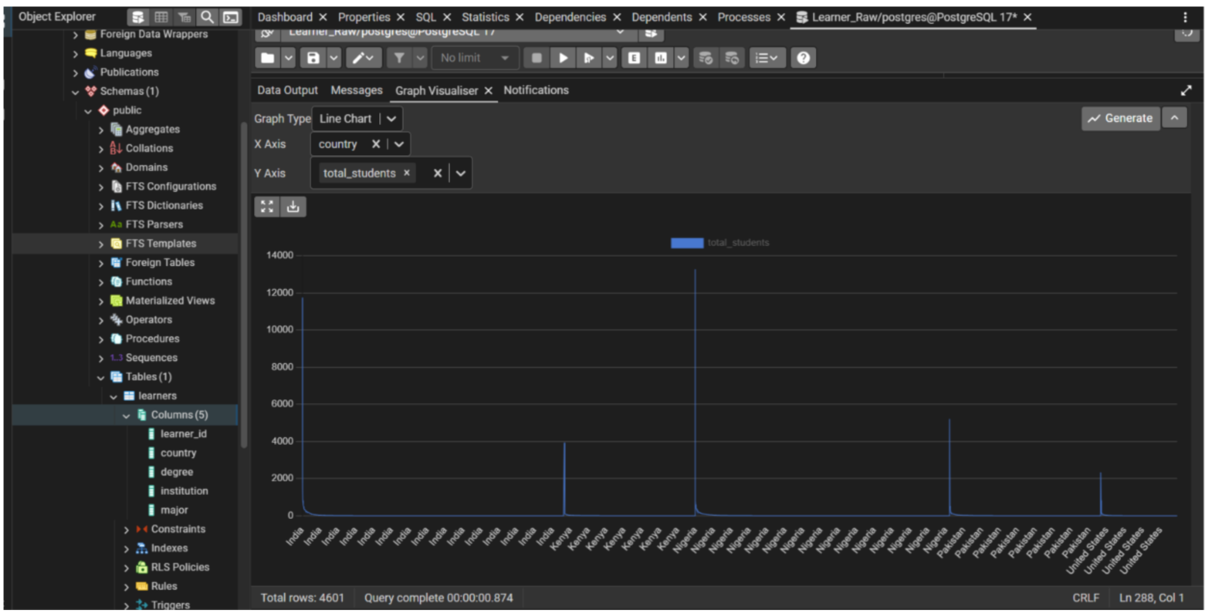



Figure: Line Chart for Major Distribution in Top 5 Countries

## Summary:

### Key Findings

1. **Dataset Overview**
   - **Rows & Columns:** 129,259 records, 5 fields (learner_id, country, degree, institution, major).
   - **Primary Key:** learner_id (all values unique).

   - **foreign keys** — single-table dataset, no relational links.
2. **Data Quality Issues**
   - **High Missing Values**
     - country: 2,275 missing
     - degree: 52,693 missing
     - institution: 52,693 missing
     - major: 52,697 missing
   - **Duplicates:** 52,692 duplicate rows detected — a major issue before further analysis.
3. **Data Types**
   - All columns are **text-based**, no numeric fields — limits certain statistical analyses (mean, median, SD, outlier detection).
4. **Visual Insights**
   - **Top 10 Countries:** Identified by learner count via bar chart.
   - **Country vs Degree** and **Degree vs Major** comparisons show category distributions.
   - **Top 5 Countries – Major Distribution** highlights concentration of fields of study by geography.
   - Numeric-based visualizations (histograms, box plots, line charts) are not meaningful here due to the absence of numerical data.

### Insights

- **Data Integrity Needs Improvement:** The large number of duplicates and missing values, particularly in academic-related columns, will significantly affect reliability if not cleaned.
- **Data Entry Standardization:** Free-text entries in degree, institution, and major likely introduce inconsistencies (spelling variations, abbreviations, formatting differences).
- **Limited Analytical Scope:** Absence of numeric variables restricts advanced statistical or trend analysis; enrichment with additional fields (e.g., enrollment date, age, scores) could expand analytical possibilities.
- **Potential Use Cases:** After cleaning, the dataset can support learner demographic analysis, degree distribution studies, and country–major mapping for educational insights.

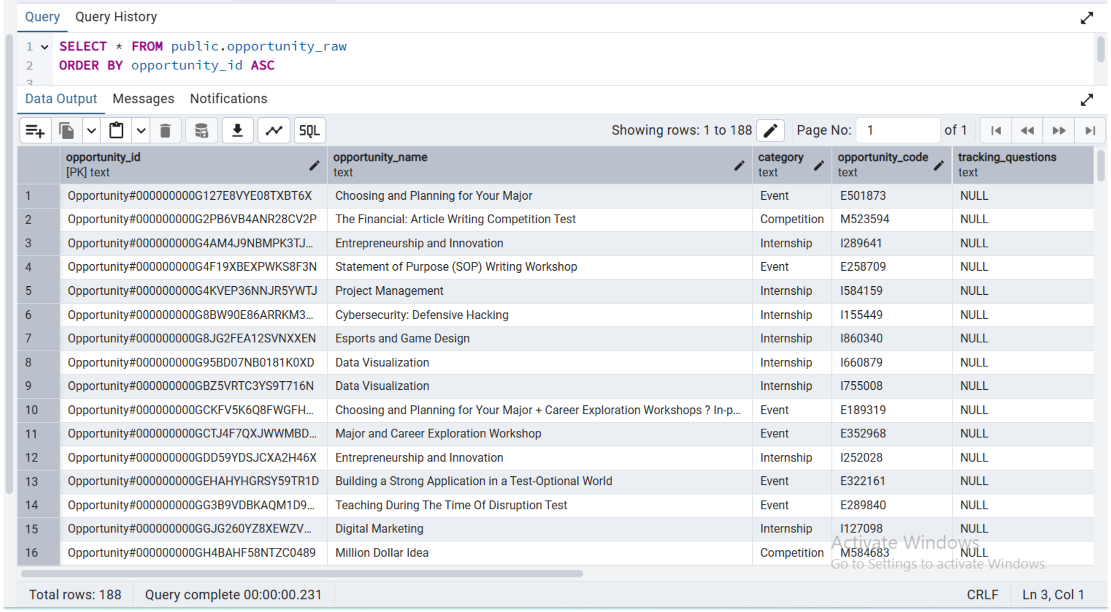
# Dataset: Opportunity Data (opp_data.csv)

## Pahse1:DATA FAMILIARIZATION

### 1. Understanding Data Sources:

The Opportunity_Raw.csv file is among six datasets supplied for the internship project. This dataset documents the professional opportunities provided to learners via Excelerate, including workshops, internships, competitions, and events. It contains 188 rows and 5 columns, featuring a variety of data types such as IDs, free-text entries, categorical information, and a JSON-like column.

### After importing "Opportunity_Raw" Dataset in PostGreSQL, Viewed All Rows of "opportunity_raw" Table:



```
SELECT * FROM public.opportunity_raw
ORDER BY opportunity_id ASC
```

| | opportunity_id [PK] text | opportunity_name text | category text | opportunity_code text | tracking_questions text |
|---|---|---|---|---|---|
| 1 | Opportunity#000000000G127E8VYE08TXBT6X | Choosing and Planning for Your Major | Event | E501873 | NULL |
| 2 | Opportunity#000000000G2PB6VB4ANR28CV2P | The Financial: Article Writing Competition Test | Competition | M523594 | NULL |
| 3 | Opportunity#000000000G4AM4J9NBMPK3TJ... | Entrepreneurship and Innovation | Internship | I289641 | NULL |
| 4 | Opportunity#000000000G4F19XBEXPWKS8F3N | Statement of Purpose (SOP) Writing Workshop | Event | E258709 | NULL |
| 5 | Opportunity#000000000G4KVEP36NNJR5YWTJ | Project Management | Internship | I584159 | NULL |
| 6 | Opportunity#000000000G8BW90E86ARRKM3... | Cybersecurity: Defensive Hacking | Internship | I155449 | NULL |
| 7 | Opportunity#000000000G8JG2FEA12SVNXXEN | Esports and Game Design | Internship | I860340 | NULL |
| 8 | Opportunity#000000000G95BD07NB0181K0XD | Data Visualization | Internship | I660879 | NULL |
| 9 | Opportunity#000000000GBZ5VRTC3YS9T716N | Data Visualization | Internship | I755008 | NULL |
| 10 | Opportunity#000000000GCKFV5K6Q8FWGFH... | Choosing and Planning for Your Major + Career Exploration Workshops ? In-p... | Event | E189319 | NULL |
| 11 | Opportunity#000000000GCTJ4F7QXJWWMBD... | Major and Career Exploration Workshop | Event | E352968 | NULL |
| 12 | Opportunity#000000000GDD59YDSJCXA2H46X | Entrepreneurship and Innovation | Internship | I252028 | NULL |
| 13 | Opportunity#000000000GEHAHYHGRSY59TR1D | Building a Strong Application in a Test-Optional World | Event | E322161 | NULL |
| 14 | Opportunity#000000000GG3B9VDBKAQM1D9... | Teaching During The Time Of Disruption Test | Event | E289840 | NULL |
| 15 | Opportunity#000000000GGJG260YZ8XEWZV... | Digital Marketing | Internship | I127098 | NULL |
| 16 | Opportunity#000000000GH4BAHF58NTZC0489 | Million Dollar Idea | Competition | M584683 | NULL |

Total rows: 188 Query complete 00:00:00.231

### 2. Review Column Meanings and Data Types:

| Column Name | Data Type | Description |
|---|---|---|
| opportunity_id | Text | Unique identifier for each opportunity (**Primary Key**) |
| opportunity_name | Text | The opportunity's title or name in free-text format |
| category | Text | Category of opportunity (such as Internship, Event, or Competition) |
| opportunity_code | Text | A unique alphanumeric code assigned to each opportunity |
| tracking_questions | Text | Holds tracking information stored as a JSON-formatted |

| | | |
|---|---|---|
| | | string |

**3. Identify Key Fields and Relationships:**

**Total Records:** 188

**Primary Key:** The primary key in this dataset is the opportunity_id column. This field serves as a unique identifier for each individual opportunity, ensuring that every record can be distinctly identified. A check of the data confirms that all 188 values in this column are unique.

**Foreign Keys:**

Based on the single file provided (Opportunity_Raw.csv), there are no foreign keys. Foreign keys are used to link data across multiple tables.

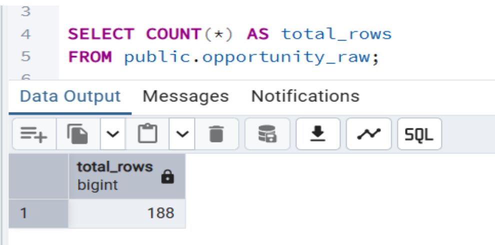
**Table Relationships:**

Since there is only one table, there are no relationships to describe.

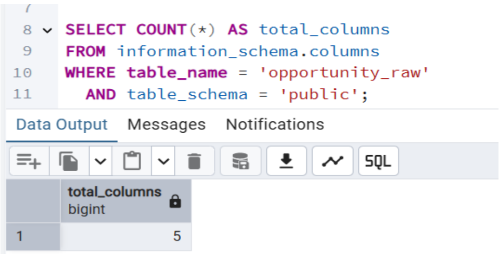
**4. Perform an Initial Data Scan:**

The Opportuniy_Raw.csv file has the following characteristics:

**Total Rows:** 188

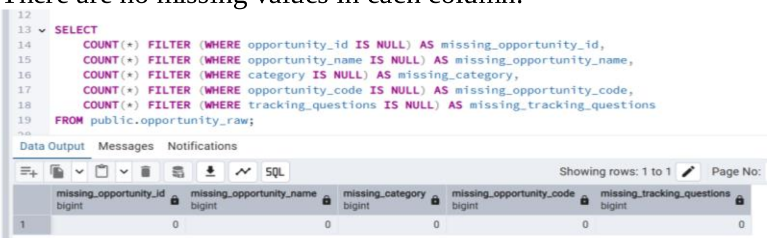



**Total Columns:** 5




**Missing Values:**

There are no missing values in each column:




**Duplicate Values:** There are duplicate records in this “Opportunity_Raw.csv” dataset.

```sql
SELECT
    (COUNT(opportunity_id) <> COUNT(DISTINCT opportunity_id)) AS opportunity_id_has_duplicates,
    (COUNT(opportunity_name) <> COUNT(DISTINCT opportunity_name)) AS opportunity_name_has_duplicates,
    (COUNT(category) <> COUNT(DISTINCT category)) AS category_has_duplicates,
    (COUNT(opportunity_code) <> COUNT(DISTINCT opportunity_code)) AS opportunity_code_has_duplicates,
    (COUNT(tracking_questions) <> COUNT(DISTINCT tracking_questions)) AS tracking_questions_has_duplicates
FROM public.opportunity_raw;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

| | opportunity_id_has_duplicates boolean | opportunity_name_has_duplicates boolean | category_has_duplicates boolean | opportunity_code_has_duplicates boolean | tracking_questions_has_duplicates boolean |
|---|---|---|---|---|---|
| 1 | false | true | true | false | true |

**NULL Values:** There are no NULL values.

```sql
SELECT
  COUNT(CASE WHEN opportunity_id IS NULL THEN 1 END) AS opportunity_id_null_count,
  COUNT(CASE WHEN opportunity_name IS NULL THEN 1 END) AS opportunity_name_null_count,
  COUNT(CASE WHEN category IS NULL THEN 1 END) AS category_null_count,
  COUNT(CASE WHEN opportunity_code IS NULL THEN 1 END) AS opportunity_code_null_count,
  COUNT(CASE WHEN tracking_questions IS NULL THEN 1 END) AS tracking_questions_null_count
FROM opportunity_raw;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1

| | opportunity_id_null_count bigint | opportunity_name_null_count bigint | category_null_count bigint | opportunity_code_null_count bigint | tracking_questions_null_count bigint |
|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 0 |

**<u>Summary of Findings:</u>**

**Key Observations:**

**Data Source:** The Opportunity_Raw.csv file appears to be a dataset containing opportunity-related records, likely exported from a sales or CRM system.

**Data Volume:**

The dataset contains 188 rows and 5 columns.

**Data Quality Issues:**

- **Missing and NULL Values:**
  There are no missing values in each columns such as opportunity_id, opportunity_name, category, opportunity_code and tracking_questions.

- **Duplicates:**
  The dataset contains duplicate rows in these columns- opportunity_name, category and tracking_questions. These is no duplicates in opportunity_id, opportunity_code.

## Exploratory Data Analysis (EDA)

**Frequencies in "Category" Column:** There are 8 types of category existing.

```sql
SELECT category, COUNT(*) AS frequency
FROM opportunity_raw
GROUP BY category
ORDER BY frequency DESC;
```

Data Output | Messages | Notifications

| | category text | frequency bigint |
|---|---|---|
| 1 | Internship | 43 |
| 2 | Event | 41 |
| 3 | Competition | 41 |
| 4 | Career | 23 |
| 5 | Course | 18 |
| 6 | Masterclass | 11 |
| 7 | Engageme... | 10 |
| 8 | category | 1 |

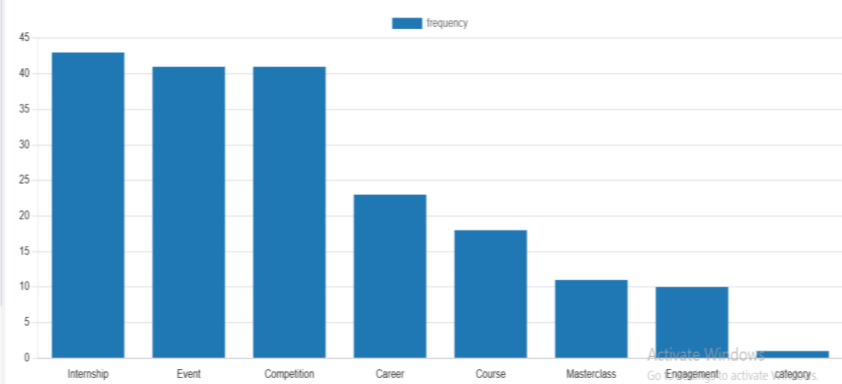



**Figure: Bar Chart of "Frequencies of each category" in 'category' column.**

## The Most Repeated Opportunity Names:

Query | Query History

```sql
SELECT opportunity_name, COUNT(*) AS repetition_count
FROM opportunity_raw
GROUP BY opportunity_name
HAVING COUNT(*) > 1
ORDER BY repetition_count DESC;
```

Data Output | Messages | Graph Visualiser | Notifications

| | opportunity_name text | repetition_count bigint |
|---|---|---|
| 1 | Project Management | 4 |
| 2 | Data Visualization | 4 |
| 3 | Cybersecurity: Defensive Hacking | 4 |
| 4 | Digital Marketing | 3 |
| 5 | Jump Start: Developing your Emotional Intelligence | 2 |
| 6 | Talent Management Associate | 2 |
| 7 | Operations Intern | 2 |
| 8 | Entrepreneurship and Innovation | 2 |
| 9 | AI Forensic Challenge | 2 |
| 10 | Million Dollar Idea | 2 |

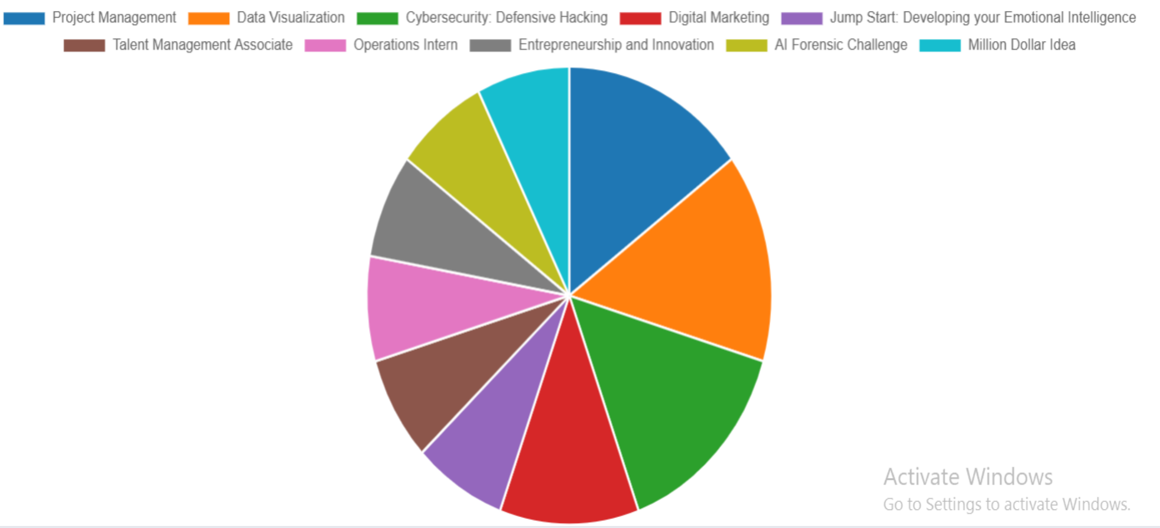
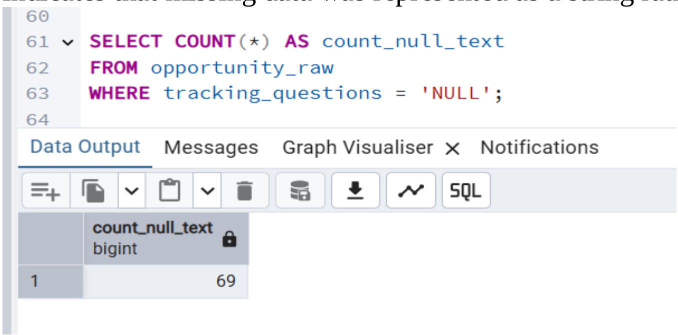



**Figure: Pie Chart of "Most repeated Opportunity Names" from 'opportunity_name'**

**'NULL' Text values in "tracking_questions" column:** Here are no actual null (missing) values in the tracking_questions column. however, some entries contain the literal text "NULL" as a placeholder. This indicates that missing data was represented as a string rather than a true null value. It is a great issue.



```
SELECT COUNT(*) AS count_null_text
FROM opportunity_raw
WHERE tracking_questions = 'NULL';
```



| | count_null_text bigint |
|---|---|
| 1 | 69 |

**Category vs Tracking Questions Availability:** This shows the number of opportunities per category that have tracking questions provided versus those missing them. It helps identify categories with more complete tracking data for better follow-up and analysis.

```
SELECT
  category,
  SUM(CASE WHEN tracking_questions IS NOT NULL AND tracking_questions <> 'NULL' THEN 1 ELSE 0 END) AS with_valid_tracking_questio
  SUM(CASE WHEN tracking_questions IS NULL OR tracking_questions = 'NULL' THEN 1 ELSE 0 END) AS without_valid_tracking_questions
FROM opportunity_raw
GROUP BY category
ORDER BY category;
```

Data Output | Messages | Graph Visualiser | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

| | category text | with_valid_tracking_questions bigint | without_valid_tracking_questions bigint |
|---|---|---|---|
| 1 | Career | 14 | 9 |
| 2 | category | 1 | 0 |
| 3 | Competition | 30 | 11 |
| 4 | Course | 16 | 2 |
| 5 | Engageme... | 7 | 3 |
| 6 | Event | 18 | 23 |
| 7 | Internship | 24 | 19 |
| 8 | Masterclass | 9 | 2 |

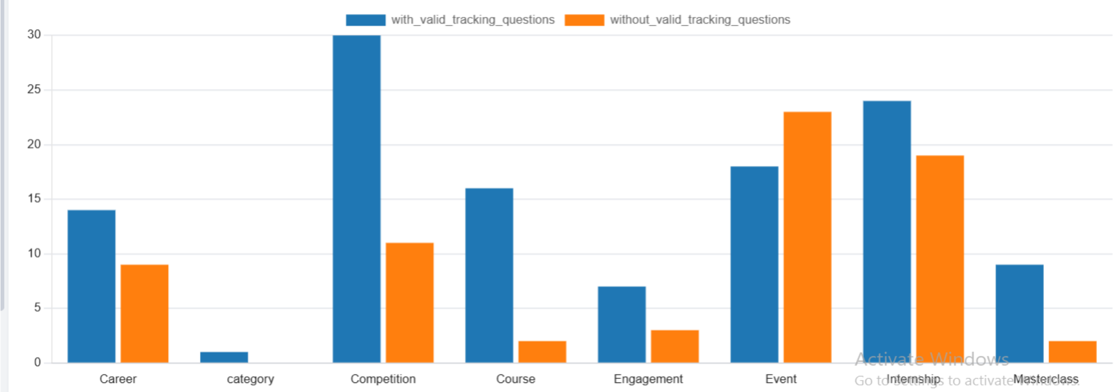



**Figure: Bar Chart of "Category vs Tracking Questions Availability"**

## Summary of Findings:

**Key Observations:**

### 1. Data Quality Issues:

- The tracking_questions column contains no true NULL (missing) values, but some entries use the literal text "NULL" and also invalid values as a placeholder for missing information.
- Other columns such as opportunity_id, opportunity_name, category, and opportunity_code are complete and non-null due to constraints. There are no duplicate opportunity records based on the primary key opportunity_id.

### 2. Category-wise Tracking Questions Completeness:

- The completeness of tracking question data varies across categories, with some categories having a higher proportion of valid entries, while others contain more placeholders or missing data.
- This highlights areas where data collection or entry processes could be improved to ensure better tracking.

**Next Steps for the "Opportunity_Raw" Dataset:**

- ✓ Clean placeholder "NULL" texts in tracking_questions by converting them to actual NULLs.
- ✓ Analyze and fill missing tracking questions to improve data completeness.
- ✓ Validate and standardize category and opportunity name entries for consistency.
- ✓ Integrate this dataset with related tables for richer opportunity insights and reporting.

## Dataset: Cohort Data (cohort_data.csv)

## Pahse1:DATA FAMILIARIZATION

This initial phase is dedicated to a thorough examination of the cohort dataset. The objective is to develop a deep understanding of its structure, content, quality, and statistical properties before proceeding to more advanced analysis.

## Data Source and Schema:

The dataset was provided as a single CSV file named CohortRaw(in).csv. It contains 639 unique records, with each record representing a distinct user cohort. A key initial data processing step involved converting the start_date and end_date columns from a non-standard numeric format (Unix timestamps in milliseconds) into a standard datetime format to enable time-based calculations and analysis.

| Column Name | Data Type (Post-Processing) | Description |
|---|---|---|
| cohort_id | Text/String | A generic, non-unique identifier for cohorts. |
| cohort_code | Text/String | A unique alphanumeric code assigned to each individual cohort. This serves as the primary key for the dataset. |
| start_date | Datetime | The specific date and time when the first user in the cohort was acquired. |
| end_date | Datetime | The specific date and time when the last user activity for that cohort was recorded. |

| | | |
|---|---|---|
| size | Integer | The total number of unique users within the cohort. |

## Data and Quality Integrity Assessment:

A series of checks were performed to validate the completeness and integrity of the dataset. The data is exceptionally clean, with no missing values or duplicate records found. The following SQL queries simulate the checks performed to verify data quality.

**Query 1: Initial Data Preview**

This query retrieves the first five records to provide a snapshot of the data's structure and content.

```
SELECT cohort_code, start_date, end_date, size FROM Cohorts LIMIT 5;
```

**Result:** The query returns the top 5 rows from the dataset, confirming the column structure and showing sample data points for each attribute, as seen in the raw CSV file.

**Query 2: Record Count Verification**

This query counts the total number of records to ensure it matches the expected count.

```
SELECT COUNT(*) AS total_records FROM Cohorts;
```

**Result:** The query confirms a total of **639** records in the dataset, which aligns perfectly with the initial data summary.

| total_records |
|---|
| 639 |

**Query 3: Duplicate Record Identification**

This query checks for duplicate cohorts by comparing the count of total records with the count of unique cohort codes.

```
SELECT COUNT(cohort_code) AS total_codes, COUNT(DISTINCT cohort_code) AS unique_codes FROM Cohorts;
```

**Result:** The counts for total and unique codes are identical, confirming that each cohort is represented by a unique cohort_code and there are no duplicates.

| total_codes | unique_codes |
|---|---|
| 639 | 639 |

**Query 4: Missing Value Analysis**

This query assesses the completeness of the data by checking for NULL values across key columns.

```
SELECT COUNT(CASE WHEN cohort_code IS NULL THEN 1 END) AS null_cohort_codes, COUNT(CASE WHEN start_date IS NULL THEN 1 END) AS null_start_dates, COUNT(CASE WHEN size IS NULL THEN 1 END) AS null_sizes FROM Cohorts;
```

**Result:** The query returns zero for all columns, verifying that the dataset has no missing values.

| null_cohort_codes | null_start_dates | null_sizes |
|---|---|---|
| 0 | 0 | 0 |

## Descriptive Statistics:

A statistical summary was generated for the numerical columns, size and the calculated duration_days. This provides critical insights into the central tendency and dispersion of the data.

| Statistic | size | duration_days |
|---|---|---|
| **count** | 639.00 | 639.00 |
| **mean** | 5,741.42 | 56.01 |
| **std** | 20,994.27 | 107.83 |
| **min** | 3.00 | 0.00 |
| **25%** | 500.00 | 29.00 |
| **50% (median)** | 800.00 | 29.00 |
| **75%** | 1,500.00 | 35.00 |
| **max** | 100,000.00 | 1,096.00 |

**Analysis of Statistics:**

- **Cohort Size (size):**
  - The **mean** cohort size is approximately 5,741, but the **median** (50th percentile) is only 800. This large difference suggests that the mean is being skewed upwards by a number of exceptionally large cohorts.
  - The **standard deviation** is extremely high (20,994.27), significantly larger than the mean itself. This indicates a massive variance in cohort sizes; they are not clustered neatly around the average.

- The range is vast, from a **minimum** of just 3 users to a **maximum** of 100,000. Furthermore, 75% of all cohorts have a size of 1,500 or less, which reinforces the idea that the largest cohorts are outliers.

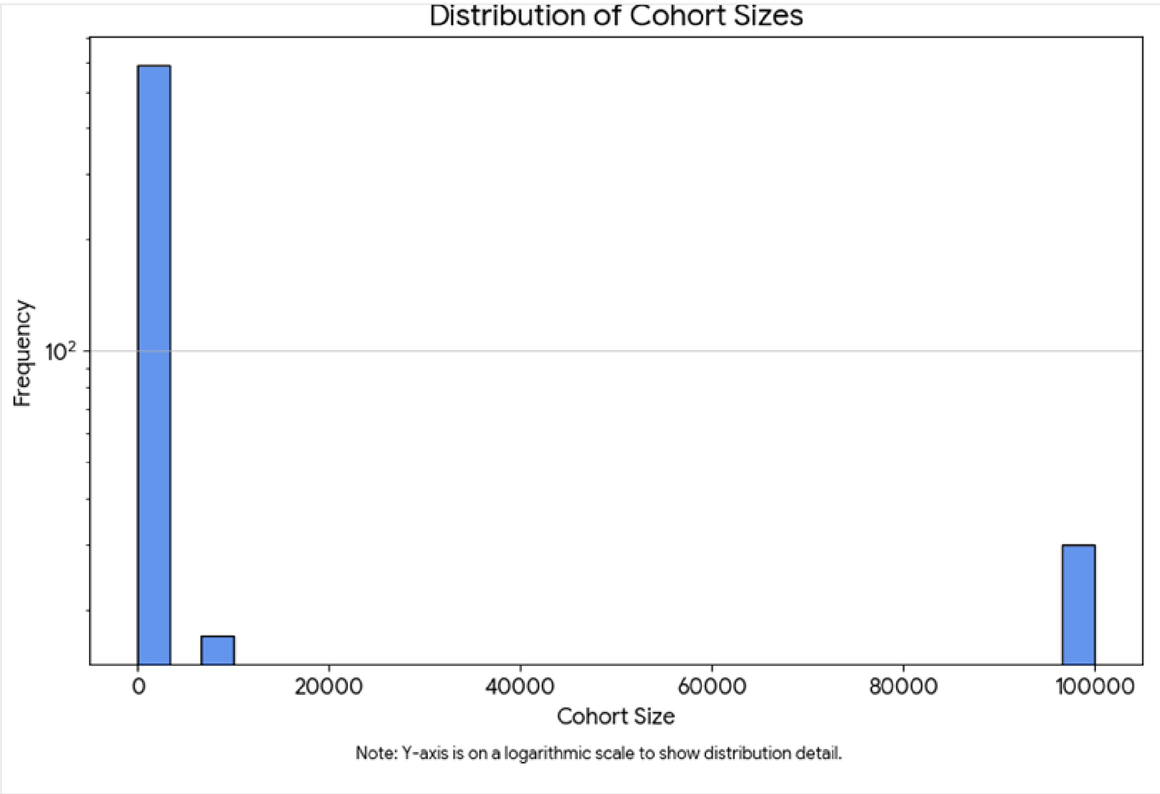
- **Cohort Duration (duration_days):**
  - Similar to size, the duration shows signs of skew. The **mean** duration is 56 days, while the **median** is much lower at 29 days.
  - The **standard deviation** of 107.83 is nearly double the mean, again pointing to high variability.
  - The range is also very wide, from a **minimum** of 0 days (where the start and end dates are the same) to a **maximum** of 1,096 days (approximately 3 years). The fact that the 25th, 50th, and 75th percentiles are all relatively close (ranging from 29 to 35 days) indicates that most cohorts have a short-to-medium duration, while a few last for a very long time.

## Phase 2: Exploratory Data Analysis (EDA)

### Visualize Data Trends:

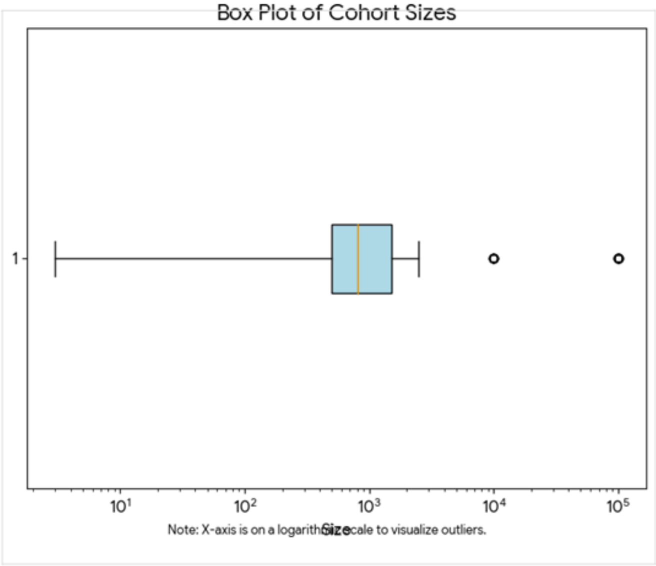
**Histogram of Cohort Sizes**

This histogram provides a stark visual confirmation of the statistical summary, revealing a strongly right-skewed distribution. The towering bar on the far left shows that the vast majority of cohorts are small, which is why the median size is only 800. In contrast, the much smaller bars to the right, including a distinct group at the 100,000 maximum, represent a few exceptionally large "hit" cohorts that pull the average size up significantly. This "hit-or-miss" pattern strongly suggests that user acquisition is not steady but is instead heavily reliant on a few key events or viral campaigns for its major growth.



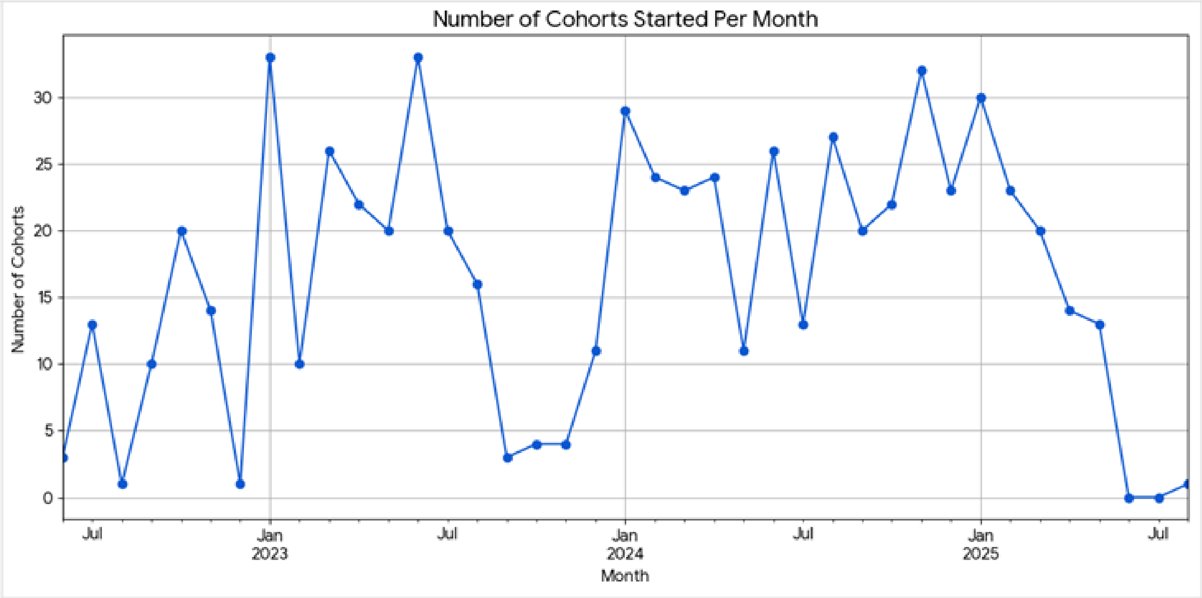

**Box Plot of Cohort Sizes**

The analysis of cohort duration reveals a mix of user engagement patterns, with notable peaks at both short and long durations. The most prominent feature is the massive peak on the left, concentrated under 50 days, which aligns with the median duration of 29 days and indicates that most cohorts are short-lived. This could signify high initial user churn or acquisitions for short-term events. Conversely, the smaller but significant clusters at much longer durations represent a highly valuable segment of engaged, long-term user groups, demonstrating loyalty and strong product-market fit.




**Line Chart of Cohorts Started Over Time**

The line graph plotting cohort size against their start date confirms that acquisition is highly volatile and driven by specific, time-bound events. Rather than a smooth progression, the graph shows a baseline of smaller, consistent acquisitions punctuated by enormous, near-vertical spikes that reach the maximum cohort size of 100,000. These significant peaks appear to be concentrated in late 2022 and again around August 2023. This visual evidence makes it clear that growth is not organic but is instead manufactured through specific efforts like product launches or major marketing campaigns, underscoring the need to analyze these peak periods to replicate success.




## Dataset:Marketing_Raw.csv

Click here to Access the Dataset: [Marketing Dataset]

**Dataset Description**
This dataset captures advertising performance, including campaign reach, engagement metrics, and costs, enabling analysis of marketing effectiveness on enrolments for the Excelerate Platform. It is a static extract representing a single day (**2023-01-01**), so time-based trend analysis is not possible.

| ***File Name*** | ***File Owner*** | ***File Type*** | ***Date Covered*** | ***Update Frequency*** | ***Limitations*** |
|---|---|---|---|---|---|
| *Marketing Campaign Data All Accounts (2023-2024)* | Excelerate | Available in both excel and csv Format | 2023-01-01 | One-time extract | Single-day snapshot; no historical trends |

**Rows and Columns:**

The marketing dataset has 13 columns and 141 rows. The details of each column are given below:

| ***Column Name*** | ***Meaning*** | ***Type*** | ***Current Data Type*** | ***Issues / Notes*** |
|---|---|---|---|---|
| *Ad Account Name* | Name/ID of the advertising account | Categorical (ID) | Text | low variation |
| *Campaign name* | Name of the campaign | Text (identifier) | Text | Two blank entries (Row 1,141) |
| *Delivery status* | Campaign's current delivery status (e.g., completed, active, inactive, achieved, not-delivering, recently-completed) | Categorical | Text | No-Issue |
| *Delivery level* | Level of delivery (only one category available campaign) | Categorical | Text | All values are `campaign` (no variation) |
| *Reach* | Number of unique users reached | Numerical (integer) | Number | Row 1 have 0 reach while:<br>• Outbound click and Outbound type is 1815 and 1310 respectively. Which is not possible.<br>• Campaign Name Column is Empty as well. |
| *Outbound clicks* | Number of clicks leading outside the platform | Numerical (integer) | Number | Two blank entries (Row 102,103) |
| *Outbound type* | Type of outbound engagement | Categorical | Text | Two blank entries (Row 102,103) |
| *Result type* | The metric being measured (e.g., Estimated ad recall lift (people), Reach, ThruPlay, Website applications submitted, Website leads) | Categorical | Text | No-Issue |
| *Results* | Count of results for the campaign objective | Numerical (integer) | Number | No-Issue |
| *Cost per result* | Cost per unit of result | Numerical (float) | Number | No-Issue |
| *Amount spent (AED)* | Total Amount spend on ad in AED or UAE Dirham | Numerical (float) | Number | No-Issue |
| *CPC (cost per link click)* | cost per click | Numerical (float) | Number | Two blank entries (Row 102,103) |
| *Reporting starts* | Date the reporting period starts | Date | Date | No-Issue |

**Database Relations:**

The dataset provided contained all the necessary attributes in a single table. Since there was no need to split the data into multiple related tables, no **primary keys** or **foreign keys** were defined. Each record is independent, and the dataset does not include unique identifiers for rows. Therefore, the entire dataset was imported as a single table for analysis.

**Exploratory Data Analysis:**

**Overview of Marketing Dataset**:

| | ad_account_name character varying (100) | campaign_name character varying (255) | delivery_status character varying (50) | delivery_level character varying (50) | reach integer | outbound_clicks integer |
|---|---|---|---|---|---|---|
| 1 | SLU | [null] | completed | campaign | 0 | 1815 |
| 2 | SLU | Digital Marketing Intern | completed | campaign | 180175 | 3378 |
| 3 | SLU | Digital Marketing Inter | inactive | campaign | 173118 | 3591 |
| 4 | SLU | Brand Awareness | inactive | campaign | 18355415 | 5431 |
| 5 | SLU | Data Analyst Associate Internship | inactive | campaign | 2448 | 111 |

| outbound_type integer | result_type character varying (100) | results integer | cost_per_result numeric (10,6) | amount_spent_aed numeric (10,2) | cpc numeric (10,6) | reporting_starts date |
|---|---|---|---|---|---|---|
| 1310 | Website applications submitted | 386 | 1.910959 | 737.63 | 0.406184 | 2023-01-01 |
| 2152 | Website applications submitted | 598 | 1.233495 | 737.63 | 0.217976 | 2023-01-01 |
| 2767 | Website applications submitted | 514 | 1.745914 | 897.40 | 0.249486 | 2023-01-01 |
| 1019 | Reach | 18355415 | 0.092997 | 1707.00 | 0.309744 | 2023-01-01 |
| 62 | Website leads | 1 | 5.430000 | 5.43 | 0.048919 | 2023-01-01 |

**Number of Rows and columns:**

| | row_count bigint | column_count bigint |
|---|---|---|
| 1 | 148 | 13 |

On initial check number of rows were 141 that means there are empty rows in the end of Marketing dataset. After removing the null rows, we get:

| | row_count bigint | column_count bigint |
|---|---|---|
| 1 | 141 | 13 |

So final number of rows are 141.

**Data Types of each Column:**

| | column_name name | data_type character varying |
|---|---|---|
| 1 | reporting_starts | date |
| 2 | outbound_type | integer |
| 3 | results | integer |
| 4 | cost_per_result | numeric |
| 5 | amount_spent_aed | numeric |
| 6 | cpc | numeric |
| 7 | reach | integer |
| 8 | outbound_clicks | integer |
| 9 | campaign_name | character varying |
| 10 | delivery_status | character varying |
| 11 | delivery_level | character varying |
| 12 | ad_account_name | character varying |
| 13 | result_type | character varying |

## Data Cleaning:

**Number of Blank Entries in each Column:**

| | reporting_starts_nulls bigint | outbound_type_nulls bigint | results_nulls bigint | cost_per_result_nulls bigint | amount_spent_aed_nulls bigint | cpc_nulls bigint | reach_nulls bigint |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 2 | 0 | 0 | 0 | 2 | 0 |

| outbound_clicks_nulls bigint | campaign_name_nulls bigint | delivery_status_nulls bigint | delivery_level_nulls bigint | ad_account_name_nulls bigint | result_type_nulls bigint |
|---|---|---|---|---|---|
| 2 | 2 | 0 | 0 | 0 | 0 |

**Issues:**

1) Two blank entries in Campaign_name Column (Row 1,141)

   Solution: Replacing blank entries by "Unnamed".

| | ad_account_name character varying (100) | campaign_name character varying (255) | delivery_status character varying (50) | delivery_level character varying (50) | reach integer | outbound_clicks integer | outbound_type integer |
|---|---|---|---|---|---|---|---|
| 1 | SLU | Unnamed | completed | campaign | 0 | 1815 | 1310 |
| 2 | SLU | Unnamed | completed | campaign | 911773 | 15679 | 10904 |

2) Multiple Blank entries in row 102 and 103.

   Solution: Replacing Blank entries by 0.

| reach integer | outbound_clicks integer | outbound_type integer | result_type character varying (100) | results integer | cost_per_result numeric (10,6) | amount_spent_aed numeric (10,2) | cpc numeric (10,6) |
|---|---|---|---|---|---|---|---|
| 1920 | 0 | 0 | Reach | 1920 | 1.286458 | 2.47 | 0.000000 |
| 17123 | 0 | 0 | Reach | 17123 | 0.057817 | 0.99 | 0.000000 |

3) No duplicate rows were found.

   Solution: No Transformation needed since there is no duplicate rows.

| | duplicate_rows bigint |
|---|---|
| 1 | 0 |

4) Row 1 have 0 reach while Outbound click and Outbound type is 1815 and 1310 respectively. Which is not possible if reach is 0.

| | ad_account_name character varying (100) | campaign_name character varying (255) | delivery_status character varying (50) | delivery_level character varying (50) | reach integer | outbound_clicks integer | outbound_type integer |
|---|---|---|---|---|---|---|---|
| 1 | SLU | Unnamed | completed | campaign | 0 | 1815 | 1310 |

   Solution: Replacing 0 by median.

| | ad_account_name character varying (100) | campaign_name character varying (255) | delivery_status character varying (50) | delivery_level character varying (50) | reach integer | outbound_clicks integer | outbound_type integer |
|---|---|---|---|---|---|---|---|
| 1 | SLU | Unnamed | completed | campaign | 148511 | 1815 | 1310 |

**Data Description:**

**Statistical description of numerical columns**

| | column_name text | min_value numeric | max_value numeric | avg_value numeric | stddev_value numeric | median_value double precision | mode_value numeric |
|---|---|---|---|---|---|---|---|
| 1 | reach | 1315 | 141835342 | 1703174.354609929078 | 12085418.0616 | 148511 | 1315 |
| 2 | outbound_clicks | 0 | 38284 | 3630.9574468085106383 | 6132.437973447126 | 1581 | 0 |
| 3 | outbound_type | 0 | 21140 | 2083.9858156028368794 | 3617.052470068127 | 848 | 0 |
| 4 | results | 1 | 142000000 | 1294443.049645390071 | 12120637.8849 | 382 | 3 |
| 5 | cost_per_result | 0.003665 | 47.089490 | 4.1639695886524823 | 4.9823117795664137 | 2.920683 | 0.003665 |
| 6 | amount_spent_aed | 0.99 | 25531.47 | 2399.9553900709219858 | 3951.404962828776 | 1222.5 | 1840.17 |
| 7 | cpc | 0.000000 | 5.597563 | 1.0316433262411348 | 0.92765934060048216490 | 0.728529 | 0.000000 |

**Number of Distinct Values in Each Categorical Column**

- Ad Account Name:

| | value character varying (100) | value_count bigint |
|---|---|---|
| 1 | SLU | 91 |
| 2 | Brand Awareness | 26 |
| 3 | RIT | 24 |

- Delivery Type:

| | value character varying (50) | value_count bigint |
|---|---|---|
| 1 | campaign | 141 |

- Delivery Status:

| | value character varying (50) | value_count bigint |
|---|---|---|
| 1 | completed | 86 |
| 2 | inactive | 37 |
| 3 | active | 10 |
| 4 | not_delivering | 4 |
| 5 | archived | 3 |
| 6 | recently_completed | 1 |

- Reach type:

| | value character varying (100) | value_count bigint |
|---|---|---|
| 1 | Website applications submitt... | 104 |
| 2 | Website leads | 15 |
| 3 | ThruPlay | 15 |
| 4 | Reach | 6 |
| 5 | Estimated ad recall lift (people) | 1 |

- Reporting Starts:

| | value date | value_count bigint |
|---|---|---|
| 1 | 2023-01-01 | 141 |

[Text Wrapping Break]**Summary**

- **Delivery Type & Reporting Start Date:**
  There is no variation in the delivery type and the reporting start date — all campaigns started on **2023-01-01**, indicating the dataset captures campaigns from a single start date.
- **Ad Accounts:**
  The dataset contains data for only **three distinct ad accounts**.
- **Delivery Status:**
  There is only **one delivery status** value present: **recently completed** and most have **active** status.
- **Reach Focus:**
  Only **one ad account** shows reach targeting **Estimated Ad Recall Lift (people).** Mostly are focused on **Website application submission.**

**Exploratory Data Analysis:**

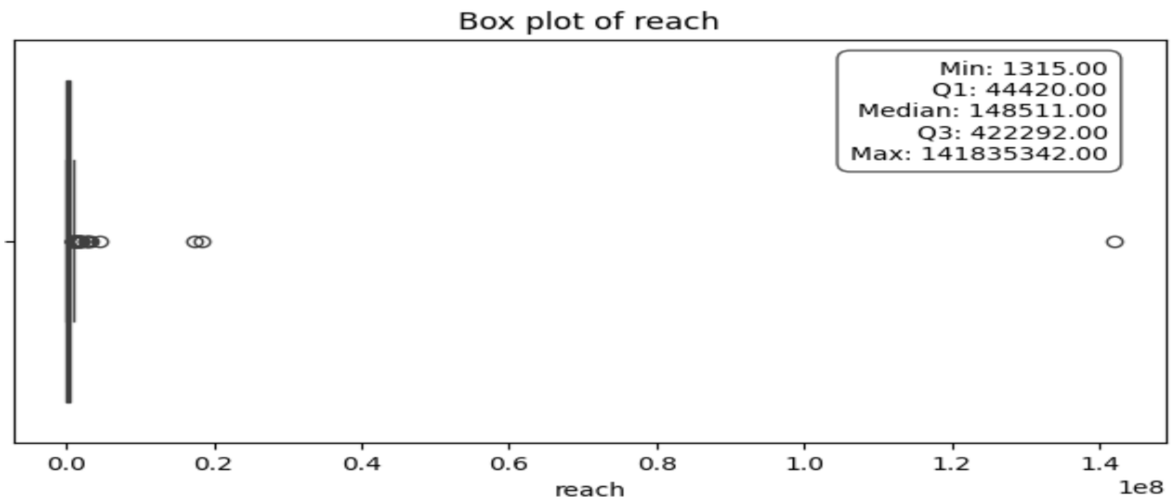
**Insights On Campaigns**

- Campaigns with most reach:

Outliers in campaign reach.

| | min_val integer | max_val integer | lower_bound double precision | upper_bound double precision | outlier_count bigint |
|---|---|---|---|---|---|
| 1 | 1315 | 141835342 | -522388 | 989100 | 26 |

All the outliers fall above 3rd Quartile.




The Top 5 campaigns with highest Reach are:

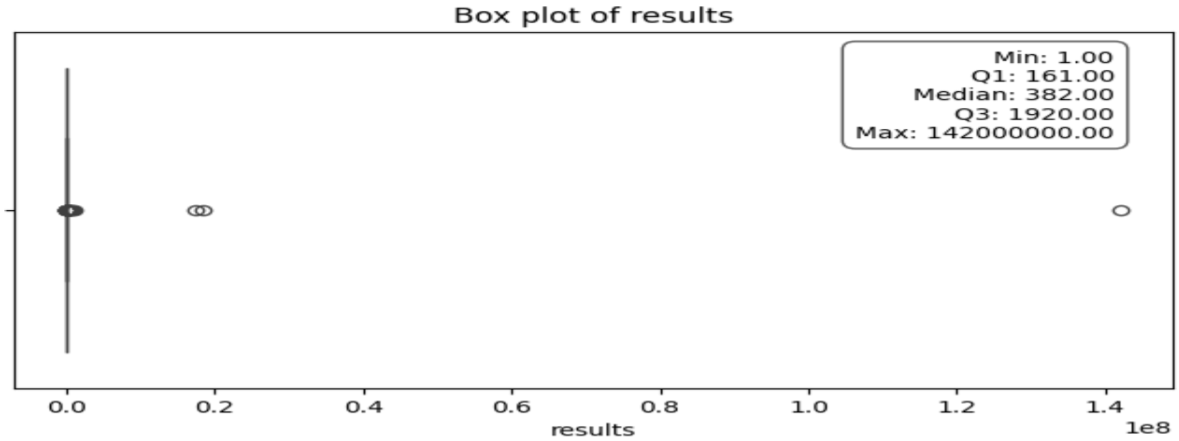
| | ad_account_name character varying (100) | campaign_name character varying (255) | delivery_status character varying (50) | delivery_level character varying (50) | reach integer |
|---|---|---|---|---|---|
| 1 | SLU | March Brand Awareness | archived | campaign | 141835342 |
| 2 | SLU | Brand Awareness | inactive | campaign | 18355415 |
| 3 | SLU | May Awareness (Reach) campaign | completed | campaign | 17331595 |
| 4 | SLU | CPR - March Ads: Website Leads Prospecting \| 18 to 35 years - Copy | inactive | campaign | 4434511 |
| 5 | Brand Awareness | FB only_AWARENESS: Excelerate_Video_Ad (with Exclusion) | completed | campaign | 3157807 |

| | outbound_clicks integer | outbound_type integer | result_type character varying (100) | results integer | cost_per_result numeric (10,6) | amount_spent_aed numeric (10,2) | cpc numeric (10,6) | reporting_starts date |
|---|---|---|---|---|---|---|---|---|
| 1 | 32289 | 5584 | Reach | 142000000 | 0.180008 | 25531.47 | 0.772043 | 2023-01-01 |
| 2 | 5431 | 1019 | Reach | 18355415 | 0.092997 | 1707.00 | 0.309744 | 2023-01-01 |
| 3 | 13469 | 2184 | Reach | 17331595 | 0.211919 | 3672.90 | 0.272026 | 2023-01-01 |
| 4 | 38284 | 16060 | Website leads | 4347 | 4.544923 | 19756.78 | 0.515385 | 2023-01-01 |
| 5 | 3228 | 439 | ThruPlay | 282986 | 0.012973 | 3671.28 | 1.135916 | 2023-01-01 |

- Campaigns with highest results.

No of Outliers in result columns.

| | min_val integer | max_val integer | lower_bound double precision | upper_bound double precision | outlier_count bigint |
|---|---|---|---|---|---|
| 1 | 1 | 142000000 | -2477.5 | 4558.5 | 24 |




All the outliers fall above $3^{rd}$ Quartile.

Top 5 campaigns with most Results are:

| | ad_account_name character varying (100) | campaign_name character varying (255) | delivery_status character varying (50) | delivery_level character varying (50) | reach integer |
|---|---|---|---|---|---|
| 1 | SLU | March Brand Awareness | archived | campaign | 141835342 |
| 2 | SLU | Brand Awareness | inactive | campaign | 18355415 |
| 3 | SLU | May Awareness (Reach) campaign | completed | campaign | 17331595 |
| 4 | Brand Awareness | Nov: FB only_AWARENESS: Excelerate_Video_Ad (with Exclusion) | completed | campaign | 389870 |
| 5 | SLU | Excelerate Brand Awareness (March Campaign) - CBO Activated - Copy | inactive | campaign | 774703 |

| | outbound_clicks integer | outbound_type integer | result_type character varying (100) | results integer | cost_per_result numeric (10,6) | amount_spent_aed numeric (10,2) | cpc numeric (10,6) | reporting_starts date |
|---|---|---|---|---|---|---|---|---|
| 1 | 32289 | 5584 | Reach | 142000000 | 0.180008 | 25531.47 | 0.772043 | 2023-01-01 |
| 2 | 5431 | 1019 | Reach | 18355415 | 0.092997 | 1707.00 | 0.309744 | 2023-01-01 |
| 3 | 13469 | 2184 | Reach | 17331595 | 0.211919 | 3672.90 | 0.272026 | 2023-01-01 |
| 4 | 1983 | 612 | ThruPlay | 1000944 | 0.003665 | 3668.72 | 1.850086 | 2023-01-01 |
| 5 | 203 | 30 | Reach | 774703 | 0.077526 | 60.06 | 0.292976 | 2023-01-01 |

Summary:

Correlation of Numerical Data:

| | col1 text | col2 text | correlation double precision |
|---|---|---|---|
| 1 | reach | outbound_clicks | 0.4419171490787199 |
| 2 | reach | results | 0.9983849830453589 |
| 3 | reach | cost_per_result | -0.09817777485316892 |
| 4 | reach | amount_spent_aed | 0.5210818709019769 |
| 5 | reach | cpc | -0.04741672312243074 |
| 6 | outbound_clicks | results | 0.4103180741954776 |
| 7 | outbound_clicks | cost_per_result | -0.126143127132948 |
| 8 | outbound_clicks | amount_spent_aed | 0.8760379699090058 |
| 9 | outbound_clicks | cpc | -0.24010180109752202 |
| 10 | results | cost_per_result | -0.0861892438324732 |
| 11 | results | amount_spent_aed | 0.4915642422330866 |
| 12 | results | cpc | -0.040168457985237195 |
| 13 | cost_per_result | amount_spent_aed | 0.038445996218503024 |
| 14 | cost_per_result | cpc | 0.5159656169519793 |
| 15 | amount_spent_aed | cpc | -0.005330000016631912 |

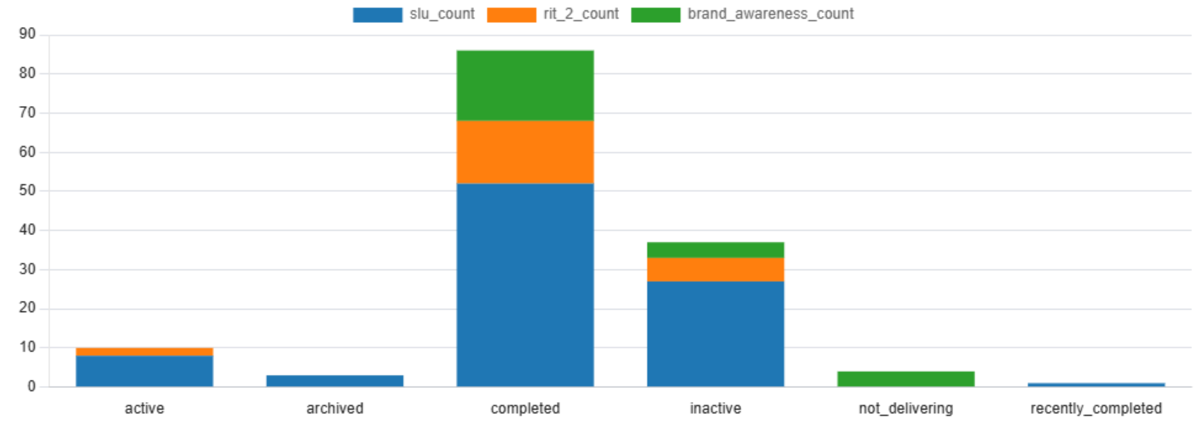
**Summary:**

- **Reach and results** are almost perfectly correlated, showing that maximizing reach is critical for better results.
- **Outbound clicks and amount spent** are very strongly correlated, indicating budget is a key driver for clicks.
- **Outbound type** strongly affects clicks and spending — this might suggest some outbound types are more effective or expensive.
- **Cost metrics (cost per result and CPC)** have moderate positive correlation, linking the cost efficiency across clicks and results.
- Negative or near-zero correlations suggest that some cost metrics are only weakly related to volume metrics like reach or results, meaning other factors impact costs independently.

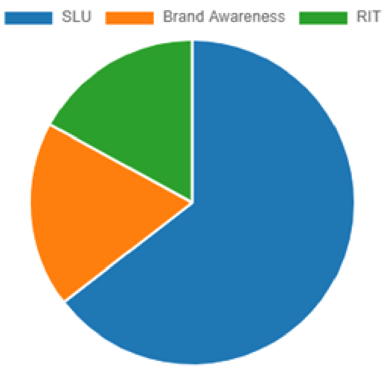
**Insights on Ad Accounts.**

- SLU has most ads campaigns while RIT and Brand Awareness ADS almost of same number but less adds then SLU

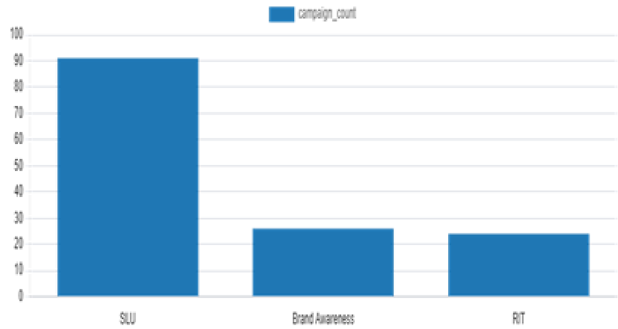
**Insights on Ad Accounts:**




- SLU has most ads campaigns while RIT and Brand Awareness ADS almost of same number but less adds then SLU




- **SLU** has the highest initial value (**4.68**) and the largest total volume, indicating it operates at the biggest scale.
- **SLU** leads by a huge margin with reach (~**2.3 million**), outbound clicks (~**4,678**), and results (~**1.96 million**), making it the most active and effective campaign category.
- **SLU** also shows the strongest efficiency or effectiveness among all categories.
- **Brand Awareness** shows moderate scale and efficiency with an initial value of **2.89**. It has strong reach **(~1.17 million)** and high results (~**143,662**), but its outbound clicks (~**2,735**) are considerably lower than SLU, indicating good engagement but comparatively less outbound activity.
- **RIT** has the smallest scale and lowest efficiency, with an initial value of **3.57**.
- It lags behind significantly with much



| | ad_account_name character varying (100) | avg_reach numeric | avg_outbound_clicks numeric | avg_results numeric |
|---|---|---|---|---|
| 1 | Brand Awareness | 1166319.884615384615 | 2734.5769230769230769 | 143661.500000000000 |
| 2 | SLU | 2293578.637362637363 | 4677.8461538461538462 | 1964590.296703296703 |
| 3 | RIT | 46150.458333333333 | 632.5833333333333333 | 148.0833333333333333 |


| | ad_account_name character varying (100) | avg_reach numeric | avg_outbound_clicks numeric | avg_results numeric |
|---|---|---|---|---|
| 1 | Brand Awareness | 1166319.884615384615 | 2734.5769230769230769 | 143661.500000000000 |
| 2 | SLU | 2293578.637362637363 | 4677.8461538461538462 | 1964590.296703296703 |
| 3 | RIT | 46150.458333333333 | 632.5833333333333333 | 148.0833333333333333 |


**Conclusion**

The analysis reveals that **SLU** is the dominant ad account, leading in campaign volume, reach, outbound clicks, and results, as well as demonstrating the highest efficiency. This highlights SLU's effectiveness in driving large-scale engagement and conversions.

In contrast, **Brand Awareness** campaigns show moderate performance with strong reach and results but lower outbound activity, suggesting room to improve engagement strategies.

**RIT** campaigns have the smallest scale and lowest impact overall, indicating they may require re-evaluation or additional support to enhance their effectiveness.

Furthermore, strong correlations between reach and results emphasize the importance of maximizing audience reach to achieve better campaign outcomes. Budget clearly influences outbound clicks, and different outbound types affect spending and engagement efficiency, suggesting tailored strategies per outbound type could optimize cost effectiveness.

Overall, focusing on scaling campaigns like SLU while improving engagement and cost efficiency in Brand Awareness and RIT could lead to more balanced and effective advertising outcomes.

## Dataset: Learner Opportunity Data (learner_opportunity_raw.csv)

**Understanding Data Sources:**

The analysis concluded that the data most likely originates from a Customer Relationship Management (CRM) system or a platform managing learner interactions. This dataset contains information about learner enrollments and includes the following columns:

enrollment_id, learner_id, assigned_cohort, apply_date, and status. This type of information is typically collected during a training or educational program's application and enrollment process to monitor learner participation, track cohort assignments, and evaluate application timelines and statuses.

| | enrollment_id character varying (100) | learner_id character varying (100) | assigned_cohort character varying (20) | apply_date timestamp without time zone | status integer |
|---|---|---|---|---|---|
| 1 | Learner#4e6f78a9-f9b2-4352-ad22-d43dc46f5ff7 | Opportunity#0000000010WCBS50CYGDX97ES4 | BAM6HBR | 2024-04-10 06:28:31.902 | 1070 |
| 2 | Learner#4e79d245-3436-4fec-9906-901a03639a… | Opportunity#0000000010WCBS50CYGDX97ES4 | BAM6HBR | 2023-11-15 03:08:17.442 | 1120 |
| 3 | Learner#4e9f5cb5-0576-4dbc-b7f5-1fae5f29b2df | Opportunity#0000000010WCBS50CYGDX97ES4 | BAM6HBR | 2024-04-06 14:07:01.322 | 1070 |
| 4 | Learner#4ea61aa9-17da-4b60-9872-359b8e1e16… | Opportunity#0000000010WCBS50CYGDX97ES4 | BT4YTCR | 2024-04-11 22:01:13.548 | 1070 |
| 5 | Learner#4eb218c7-467a-470a-9e3e-a2b7bc649e… | Opportunity#0000000010WCBS50CYGDX97ES4 | BT4YTCR | 2024-10-22 15:44:13.402 | 1120 |
| 6 | Learner#4ec728db-7d09-4a8e-b1ab-dab3011a55… | Opportunity#0000000010WCBS50CYGDX97ES4 | BT4YTCR | 2024-04-12 07:00:26.574 | 1070 |
| 7 | Learner#4edc150c-ea73-4144-993f-77ca8124de… | Opportunity#0000000010WCBS50CYGDX97ES4 | BT4YTCR | 2024-11-24 11:07:11.742 | 1070 |
| 8 | Learner#4ef3715a-e420-4296-908c-eab9e5b797… | Opportunity#0000000010WCBS50CYGDX97ES4 | BC69M2K | 2025-02-18 12:41:51.125 | 1070 |
| 9 | Learner#4ef49684-e9b0-40a9-b07e-07bfa78bdbc5 | Opportunity#0000000010WCBS50CYGDX97ES4 | BGRQZ2N | 2024-10-08 09:50:06.608 | 1120 |
| 10 | Learner#4f027693-d86f-4d65-a0a1-6342c8c097… | Opportunity#0000000010WCBS50CYGDX97ES4 | BAM6HBR | 2024-04-04 15:15:45.184 | 1070 |
| 11 | Learner#4f0328f9-144e-4fe7-a4a0-604517a7655f | Opportunity#0000000010WCBS50CYGDX97ES4 | BGRQZ2N | 2025-01-31 16:49:17.139 | 1070 |
| 12 | Learner#4f19b0d8-d5f4-463b-b78c-ea2b74db54… | Opportunity#0000000010WCBS50CYGDX97ES4 | BAM6HBR | 2024-04-10 23:18:43.408 | 1070 |
| 13 | Learner#63cea3a8-615f-4cc9-b230-096d500923… | Opportunity#000000000G127E8VYE08TXBT6X | B880483 | 2022-08-19 01:53:08.792 | 1120 |
| 14 | Learner#4f253c7f-3f78-4ae3-ac59-24b9bb138fd3 | Opportunity#0000000010WCBS50CYGDX97ES4 | BAM6HBR | 2024-04-05 02:04:06.91 | 1070 |
| 15 | Learner#4f27c524-cb7b-4cf3-aa5f-c190f773ea18 | Opportunity#0000000010WCBS50CYGDX97ES4 | BGRQZ2N | 2024-09-24 23:55:10.584 | 1070 |
| 16 | Learner#4f32e611-9b57-4bea-9b12-f6f5829f544b | Opportunity#0000000010WCBS50CYGDX97ES4 | BAM6HBR | 2024-03-30 05:41:03.93 | 1120 |
| 17 | Learner#4f347f56-ea7c-4b1f-a67d-20287f504ed2 | Opportunity#0000000010WCBS50CYGDX97ES4 | BAM6HBR | 2024-04-04 10:34:14.417 | 1070 |
| 18 | Learner#4f4ec266-5573-4463-8771-f178de3a0d… | Opportunity#0000000010WCBS50CYGDX97ES4 | BAM6HBR | 2024-03-30 05:40:08.938 | 1070 |

Review Column Meanings and Data Types:

| | |
|---|---|
| **Enrollment_id** | A unique identifier for each enrollment record. The data type is a string (object), which is appropriate for maintaining uniqueness. |
| **learner_id** | A unique identifier for each learner. The data type is a string, suitable for tracking individual learners across systems. |

| | |
|---|---|
| **assigned_cohort** | Indicates the cohort or batch to which the learner is assigned. This is stored as a string, which is appropriate for labeling group names. |
| **apply_date** | The date when the learner applied. It is stored as a string, but it represents date information and should ideally be converted to a date format for better analysis. |
| **status** | A numeric code representing the learner's enrollment status. The data type is an integer, which is suitable if the status is based on predefined numeric codes |

# Identify Key Fields:

## Primary Key:

### enrollment_id

This is the unique identifier for each enrollment record and serves as the **primary key** of the table.
**Reason:** it contains unique values for each enrollment.

## Foreign Keys:

### learner_id

This likely serves as a **foreign key** referencing a **learners** table that contains detailed information about each learner (e.g., name, contact info, demographics).
**Reason:** It connects each enrollment to a specific learner.

### assigned_cohort

This may be a **foreign key** referring to a **cohorts** table that stores details about each cohort (e.g., cohort name, start date, capacity).
**Reason:** It groups learners based on their assigned training batch.

### status

This could be a **foreign key** pointing to a **status lookup table** containing human- readable meanings for each status code (e.g., 1010 = Applied, 1020 = Shortlisted). **Reason:** Numeric codes are used for efficient storage, and their meanings are usually maintained in a separate reference table.

# Perform an initial data scan:

The LearnerOpportunity dataset has the following characteristics: Rows and

**Columns**: The dataset has 113602 rows and 5 columns

| | count bigint |
|---|---|
| 1 | 113602 |

Data Output Messages Notifications

| | column_name name | data_type character varying |
|---|---|---|
| 1 | apply_date | timestamp without time zone |
| 2 | status | integer |
| 3 | enrollment_id | character varying |
| 4 | learner_id | character varying |
| 5 | assigned_cohort | character varying |

**Missing Values:** assigned_cohort: 13318

missing values apply_date: 188 missing

values. status: 186 missing values.

Data Output Messages Notifications

Showing rows: 1 to 1

| | enrollment_id_nulls bigint | learner_id_nulls bigint | assigned_cohort_nulls bigint | apply_date_nulls bigint | status_nulls bigint |
|---|---|---|---|---|---|
| 1 | 0 | 0 | 13318 | 188 | 186 |

## Summary of Findings:

Key observations:

- **Data Source**: The LearnerOpportunity dataset most likely originates from a Customer Relationship Management (CRM) system or a platform managing learner interactions.
- **Data Volume:** The dataset contains 113602 rows and 5 columns.
- **Data Quality Issues:**
  - ♣ **Missing Values:** The dataset contains missing values in three columns: assigned_cohort (13,318), apply_date (188), and status (186), indicating potential issues in data completeness.
  - ♣ **Duplicates:** There are no completely duplicate rows in the dataset.

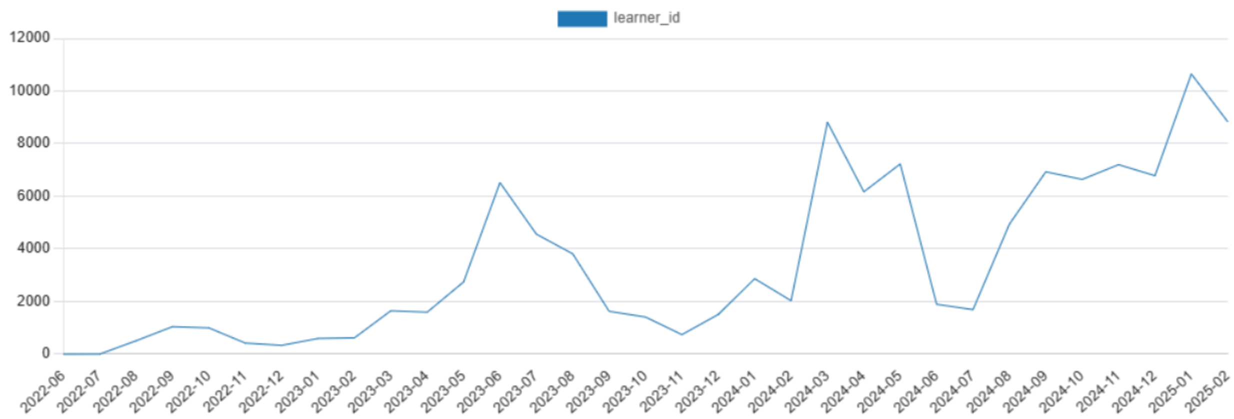
# DATA VISUALISATION:

Monthly Learner Enrollment Trends (2022–2025):

```
SELECT TO_CHAR(apply_date, 'YYYY-MM') AS year_month,
       COUNT(*) AS learner_id
FROM enrollment
WHERE apply_date IS NOT NULL
GROUP BY TO_CHAR(apply_date, 'YYYY-MM')
ORDER BY year_month;
```

:a Output   Messages   Graph Visualiser ×   Notifications

| year_month text | learner_id bigint |
|---|---|
| 2022-06 | 1 |
| 2022-07 | 5 |
| 2022-08 | 511 |
| 2022-09 | 1046 |
| 2022-10 | 1000 |
| 2022-11 | 424 |
| 2022-12 | 337 |
| 2023-01 | 598 |
| 2023-02 | 621 |
| 2023-03 | 1650 |
| 2023-04 | 1597 |



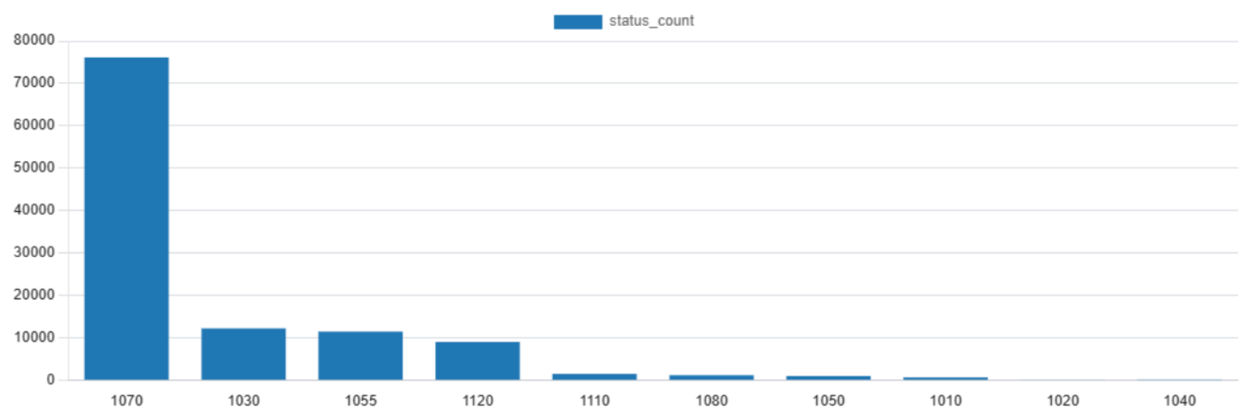

The line graph shows monthly learner enrollments from mid-2022 to early 2025, revealing a clear upward trend with cyclical fluctuations. Numbers were low in 2022 and early 2023, followed by notable peaks in June 2023 and February 2024. From mid-2024, enrollments grew steadily, reaching the highest point of over 10,000 learners in January 2025 before slightly declining in February 2025, indicating both seasonal patterns and sustained growth.

## Status Frequency:

```sql
SELECT TO_CHAR(apply_date, 'YYYY-MM') AS year_month,
       COUNT(*) AS learner_id
FROM enrollment
WHERE apply_date IS NOT NULL
GROUP BY TO_CHAR(apply_date, 'YYYY-MM')
ORDER BY year_month;
```

a Output Messages Graph Visualiser Notifications

| year_month text | learner_id bigint |
|---|---|
| 2022-06 | 1 |
| 2022-07 | 5 |
| 2022-08 | 511 |
| 2022-09 | 1046 |
| 2022-10 | 1000 |
| 2022-11 | 424 |
| 2022-12 | 337 |
| 2023-01 | 598 |
| 2023-02 | 621 |
| 2023-03 | 1650 |
| 2023-04 | 1597 |




One limitation of the database is that the status and assigned_cohort columns use codes instead of clear labels. This means the values cannot be easily understood without looking at a separate guide or codebook. As a result, graphs made from these columns only show the codes, which makes it hard for viewers to understand the trends or patterns. This reduces how easily the data can be used for analysis and reporting.

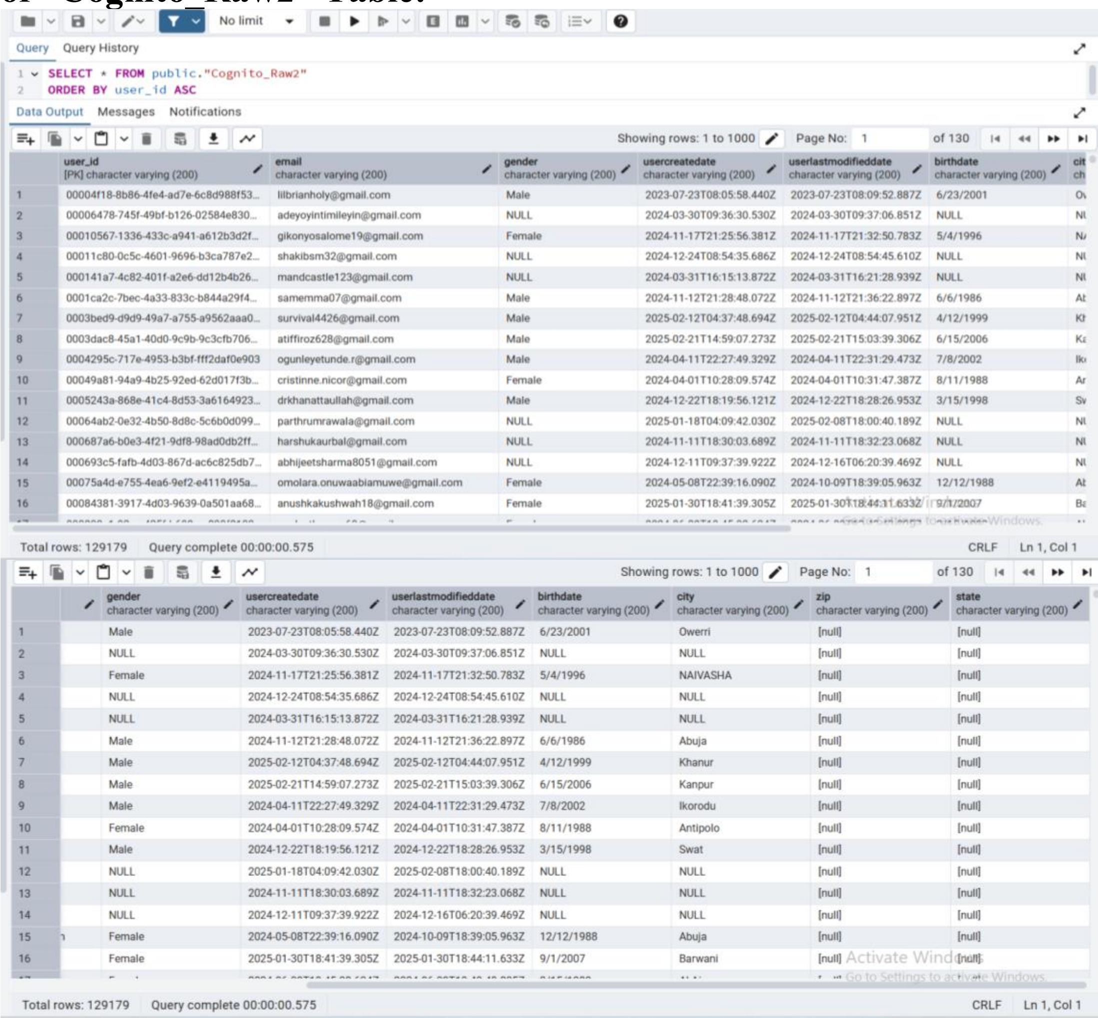
**Dataset: Cognito Data (cognito_raw.csv)**

## Phase 1 DATA FAMILIARIZATION

### 1. Understanding Data Sources:

The Cognito_Raw2.csv file is one of among six datasets provided in the “Excelerate Data Visualization Associate Internship” project. This dataset contains Cognito data including columns for user_id, email, gender, usercreatedate, userlastmodifieddate, birthdate, city, zip, state. This type of data is typically generated from a user authentication or identity management system, suggesting it was likely exported from an identity provider platform or user account database. A considerable amount of missing and inconsistent data is present, particularly in the birthdate, gender, city, and zip fields. Additionally, some columns such as gender and city use free-text input, which may lead to irregularities, non-standard entries, and challenges in analysis.

### After importing “Cognito_Raw2” Dataset in PostGreSQL, Viewed All Rows of “Cognito_Raw2” Table:



```
SELECT * FROM public."Cognito_Raw2"
ORDER BY user_id ASC
```

Showing rows: 1 to 1000 Page No: 1 of 130

| | user_id [PK] character varying (200) | email character varying (200) | gender character varying (200) | usercreatedate character varying (200) | userlastmodifieddate character varying (200) | birthdate character varying (200) |
|---|---|---|---|---|---|---|
| 1 | 00004f18-8b86-4fe4-ad7e-6c8d988f53... | lilbrianholy@gmail.com | Male | 2023-07-23T08:05:58.440Z | 2023-07-23T08:09:52.887Z | 6/23/2001 |
| 2 | 00006478-745f-49bf-b126-02584e830... | adeyoyintimileyin@gmail.com | NULL | 2024-03-30T09:36:30.530Z | 2024-03-30T09:37:06.851Z | NULL |
| 3 | 00010567-1336-433c-a941-a612b3d2f... | gikonyosalome19@gmail.com | Female | 2024-11-17T21:25:56.381Z | 2024-11-17T21:32:50.783Z | 5/4/1996 |
| 4 | 00011c80-0c5c-4601-9696-b3ca787e2... | shakibsm32@gmail.com | NULL | 2024-12-24T08:54:35.686Z | 2024-12-24T08:54:45.610Z | NULL |
| 5 | 000141a7-4c82-401f-a2e6-dd12b4b26... | mandcastle123@gmail.com | NULL | 2024-03-31T16:15:13.872Z | 2024-03-31T16:21:28.939Z | NULL |
| 6 | 0001ca2c-7bec-4a33-833c-b844a29f4... | samemma07@gmail.com | Male | 2024-11-12T21:28:48.072Z | 2024-11-12T21:36:22.897Z | 6/6/1986 |
| 7 | 0003bed9-d9d9-49a7-a755-a9562aaa0... | survival4426@gmail.com | Male | 2025-02-12T04:37:48.694Z | 2025-02-12T04:44:07.951Z | 4/12/1999 |
| 8 | 0003dac8-45a1-40d0-9c9b-9c3cfb706... | atiffiroz628@gmail.com | Male | 2025-02-21T14:59:07.273Z | 2025-02-21T15:03:39.306Z | 6/15/2006 |
| 9 | 0004295c-717e-4953-b3bf-fff2daf0e903 | ogunleyetunde.r@gmail.com | Male | 2024-04-11T22:27:49.329Z | 2024-04-11T22:31:29.473Z | 7/8/2002 |
| 10 | 00049a81-94a9-4b25-92ed-62d017f3b... | cristinne.nicor@gmail.com | Female | 2024-04-01T10:28:09.574Z | 2024-04-01T10:31:47.387Z | 8/11/1988 |
| 11 | 0005243a-868e-41c4-8d53-3a6164923... | drkhanattaullah@gmail.com | Male | 2024-12-22T18:19:56.121Z | 2024-12-22T18:28:26.953Z | 3/15/1998 |
| 12 | 00064ab2-0e32-4b50-8d8c-5c6b0d099... | parthrumrawala@gmail.com | NULL | 2025-01-18T04:09:42.030Z | 2025-02-08T18:00:40.189Z | NULL |
| 13 | 000687a6-b0e3-4f21-9df8-98ad0db2ff... | harshukaurbal@gmail.com | NULL | 2024-11-11T18:30:03.689Z | 2024-11-11T18:32:23.068Z | NULL |
| 14 | 000693c5-fafb-4d03-867d-ac6c825db7... | abhijeetsharma8051@gmail.com | NULL | 2024-12-11T09:37:39.922Z | 2024-12-16T06:20:39.469Z | NULL |
| 15 | 00075a4d-e755-4ea6-9ef2-e4119495a... | omolara.onuwaabiamuwe@gmail.com | Female | 2024-05-08T22:39:16.090Z | 2024-10-09T18:39:05.963Z | 12/12/1988 |
| 16 | 00084381-3917-4d03-9639-0a501aa68... | anushkakushwah18@gmail.com | Female | 2025-01-30T18:41:39.305Z | 2025-01-30T18:44:11.633Z | 9/1/2007 |

Total rows: 129179 Query complete 00:00:00.575 CRLF Ln 1, Col 1

Showing rows: 1 to 1000 Page No: 1 of 130

| | gender character varying (200) | usercreatedate character varying (200) | userlastmodifieddate character varying (200) | birthdate character varying (200) | city character varying (200) | zip character varying (200) | state character varying (200) |
|---|---|---|---|---|---|---|---|
| 1 | Male | 2023-07-23T08:05:58.440Z | 2023-07-23T08:09:52.887Z | 6/23/2001 | Owerri | [null] | [null] |
| 2 | NULL | 2024-03-30T09:36:30.530Z | 2024-03-30T09:37:06.851Z | NULL | NULL | [null] | [null] |
| 3 | Female | 2024-11-17T21:25:56.381Z | 2024-11-17T21:32:50.783Z | 5/4/1996 | NAIVASHA | [null] | [null] |
| 4 | NULL | 2024-12-24T08:54:35.686Z | 2024-12-24T08:54:45.610Z | NULL | NULL | [null] | [null] |
| 5 | NULL | 2024-03-31T16:15:13.872Z | 2024-03-31T16:21:28.939Z | NULL | NULL | [null] | [null] |
| 6 | Male | 2024-11-12T21:28:48.072Z | 2024-11-12T21:36:22.897Z | 6/6/1986 | Abuja | [null] | [null] |
| 7 | Male | 2025-02-12T04:37:48.694Z | 2025-02-12T04:44:07.951Z | 4/12/1999 | Khanur | [null] | [null] |
| 8 | Male | 2025-02-21T14:59:07.273Z | 2025-02-21T15:03:39.306Z | 6/15/2006 | Kanpur | [null] | [null] |
| 9 | Male | 2024-04-11T22:27:49.329Z | 2024-04-11T22:31:29.473Z | 7/8/2002 | Ikorodu | [null] | [null] |
| 10 | Female | 2024-04-01T10:28:09.574Z | 2024-04-01T10:31:47.387Z | 8/11/1988 | Antipolo | [null] | [null] |
| 11 | Male | 2024-12-22T18:19:56.121Z | 2024-12-22T18:28:26.953Z | 3/15/1998 | Swat | [null] | [null] |
| 12 | NULL | 2025-01-18T04:09:42.030Z | 2025-02-08T18:00:40.189Z | NULL | NULL | [null] | [null] |
| 13 | NULL | 2024-11-11T18:30:03.689Z | 2024-11-11T18:32:23.068Z | NULL | NULL | [null] | [null] |
| 14 | NULL | 2024-12-11T09:37:39.922Z | 2024-12-16T06:20:39.469Z | NULL | NULL | [null] | [null] |
| 15 | Female | 2024-05-08T22:39:16.090Z | 2024-10-09T18:39:05.963Z | 12/12/1988 | Abuja | [null] | [null] |
| 16 | Female | 2025-01-30T18:41:39.305Z | 2025-01-30T18:44:11.633Z | 9/1/2007 | Barwani | [null] | [null] |

Total rows: 129179 Query complete 00:00:00.575 CRLF Ln 1, Col 1

### 2. Review Column Meanings and Data Types:

| Column Name | Data Type | Description |
|---|---|---|
| user_id | Character varying (200) | Unique identification for each user |

| | | **(Primary Key)** |
|---|---|---|
| email | Character varying (200) | User's email address |
| gender | Character varying (200) | Gender identity (Male, Female, Other, etc.) |
| usercreatedate | Character varying (200) | Profile creation timestamp |
| userlastmodifieddate | Character varying (200) | Last profile modification timestamp |
| birthdate | Character varying (200) | User's date of birth |
| city | Character varying (200) | City from user ID |
| zip | Character varying (200) | Zip or postal code |
| state | Character varying (200) | State or region |

## 3. <u>Identify Key Fields and Relationships:</u>

**Total Records:** 129,179

**Primary Key:** The primary key in this dataset is the user_id column. This field serves as a unique identifier for each individual user, ensuring that every record can be distinctly identified. A check of the data confirms that all 129,179 values in this column are unique.

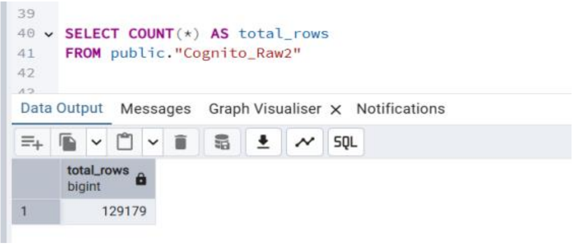
**Foreign Keys:**
Based on the single file provided (Cognito_Raw2.csv), there are no foreign keys. Foreign keys are used to link data across multiple tables.

**Table Relationships:**
Since there is only one table, there are no relationships to describe.

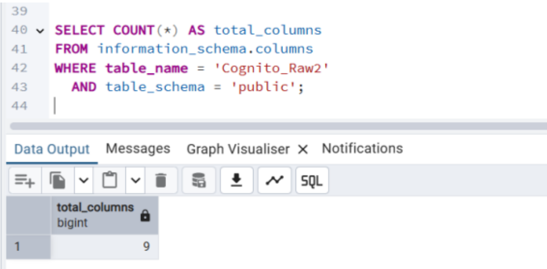
## 4. <u>Perform an Initial Data Scan</u>:

The Cognito_Raw2.csv file has the following characteristics:

**Total Rows:** 129179




**Total Columns:** 9

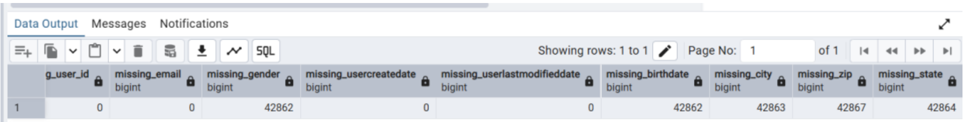


```sql
SELECT COUNT(*) AS total_columns
FROM information_schema.columns
WHERE table_name = 'Cognito_Raw2'
  AND table_schema = 'public';
```

| | total_columns bigint |
|---|---|
| 1 | 9 |

## Missing Values:

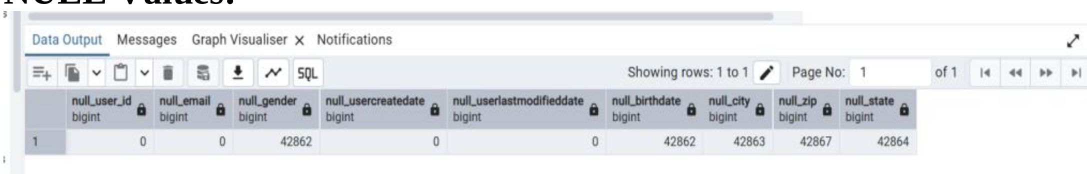
There are significant missing values in several columns:
Gender :42862
Birthdate: 42862
City : 42863
Zip: 42867
State: 42864



| | g_user_id bigint | missing_email bigint | missing_gender bigint | missing_usercreatedate bigint | missing_userlastmodifieddate bigint | missing_birthdate bigint | missing_city bigint | missing_zip bigint | missing_state bigint |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 42862 | 0 | 0 | 42862 | 42863 | 42867 | 42864 |

**Duplicate Values:** There is no duplicate records in this "Cognito_Raw2.csv" dataset.

## NULL Values:



| | null_user_id bigint | null_email bigint | null_gender bigint | null_usercreatedate bigint | null_userlastmodifieddate bigint | null_birthdate bigint | null_city bigint | null_zip bigint | null_state bigint |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 42862 | 0 | 0 | 42862 | 42863 | 42867 | 42864 |

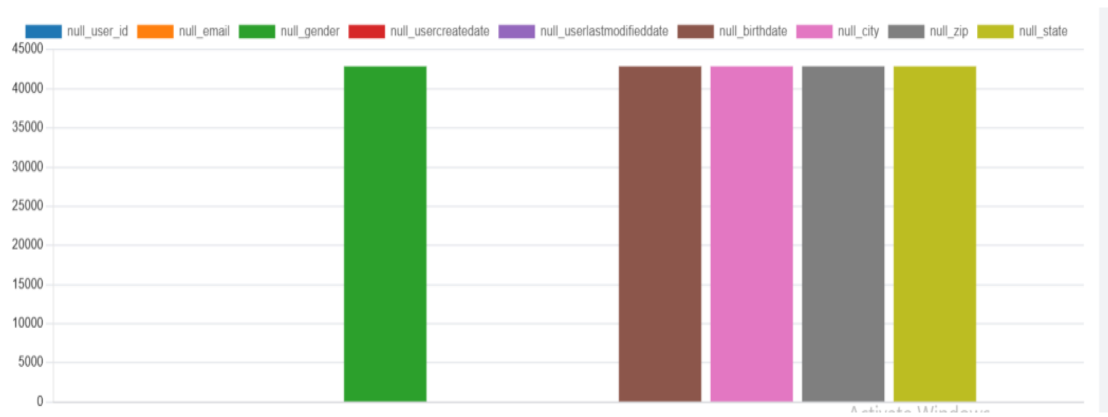



**Figure: Bar Chart of 'NULL' values in each column.**

## Summary of Findings:

### Key Observations:

1. **Data Source:**
   The Cognito_Raw2.csv file appears to be a user profile dataset, likely exported from a user identity or authentication management system.
2. **Data Volume:**

The dataset contains 129,179 rows and 9 columns.

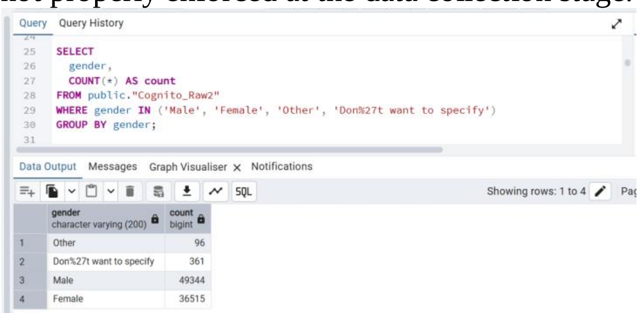
## Data Quality Issues:

- **Missing Values:**
  There's a high number of missing values, particularly in the gender, birthdate, city, zip, and state columns.

- **Duplicates:**
  The dataset contains no duplicate rows. This indicates that each user record is unique and no redundancy was detected.

## Exploratory Data Analysis (EDA)

**“Gender” Column :** The gender column in the ‘Cognito_Raw2’ table contains some values that resemble URLs or contain non-standard entries, which suggests that input validation was not properly enforced at the data collection stage.



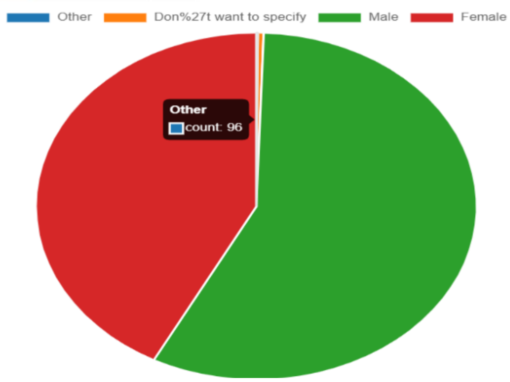




**Figure: Pie Chart of ‘gender’ column.**

**Top 10 cities & states from “City” and “State” Column-Overview:**

Query | Query History

```sql
SELECT
  city,
  COUNT(*) AS user_count
FROM public."Cognito_Raw2"
WHERE city IS NOT NULL
  AND city != 'NULL'
GROUP BY city
ORDER BY user_count DESC
LIMIT 10;
```

Data Output | Messages | Graph Visualiser | Notifications

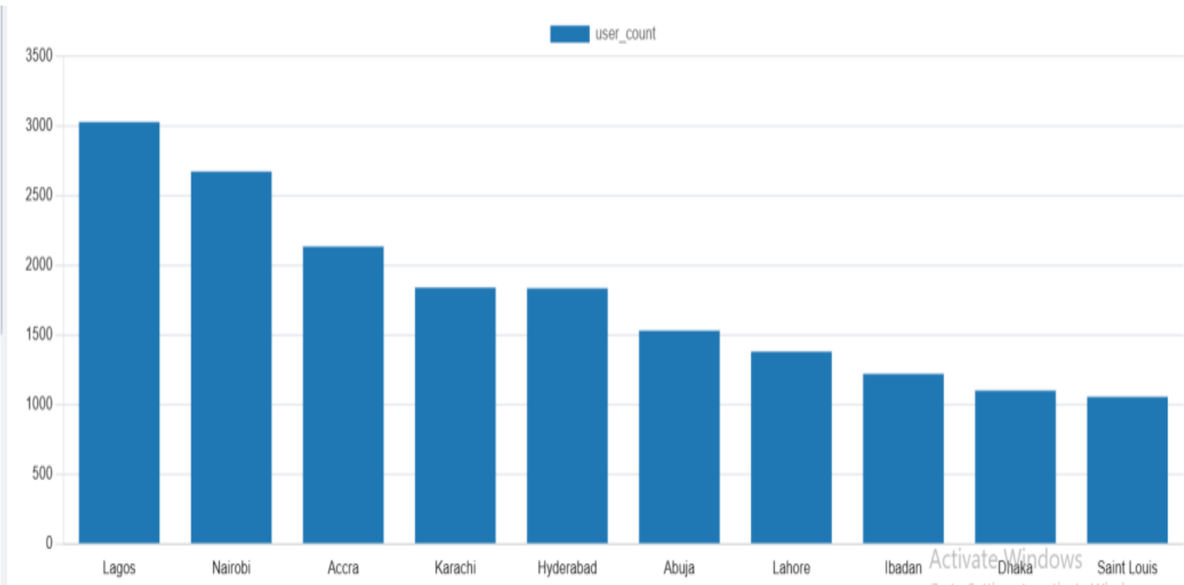
| | city character varying (200) | user_count bigint |
|---|---|---|
| 1 | Lagos | 3031 |
| 2 | Nairobi | 2675 |
| 3 | Accra | 2136 |
| 4 | Karachi | 1841 |
| 5 | Hyderabad | 1836 |
| 6 | Abuja | 1532 |
| 7 | Lahore | 1381 |
| 8 | Ibadan | 1221 |
| 9 | Dhaka | 1101 |
| 10 | Saint Louis | 1056 |

Query | Query History

```sql
SELECT
  state,
  COUNT(*) AS user_count
FROM public."Cognito_Raw2"
WHERE state IS NOT NULL
  AND state != 'NULL'
GROUP BY state
ORDER BY user_count DESC
LIMIT 10;
```

Data Output | Messages | Graph Visualiser | Notifications

| | state character varying (200) | user_count bigint |
|---|---|---|
| 1 | Lagos | 6154 |
| 2 | Punjab | 3677 |
| 3 | Maharashtra | 3487 |
| 4 | Telangana | 2434 |
| 5 | Sindh | 2040 |
| 6 | Karnataka | 1653 |
| 7 | Missouri | 1648 |
| 8 | Andhra Pradesh | 1616 |
| 9 | Nairobi | 1507 |
| 10 | Uttar Pradesh | 1333 |



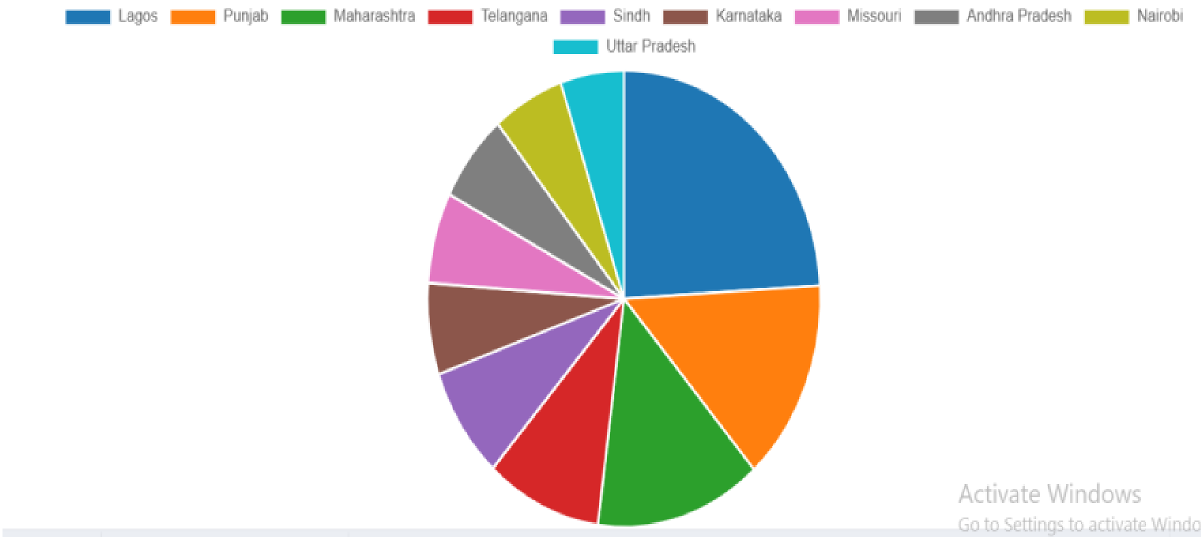
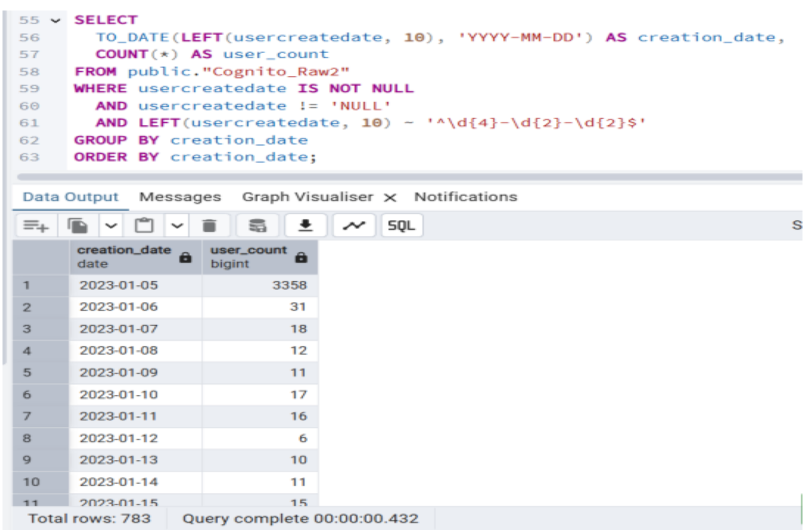



**Figure: Bar Chart of "City" Column & Pie Chart of 'State' column.**

**Daily User Account Creations from 'usercreatedate' column-Overview:**



```sql
SELECT
  TO_DATE(LEFT(usercreatedate, 10), 'YYYY-MM-DD') AS creation_date,
  COUNT(*) AS user_count
FROM public."Cognito_Raw2"
WHERE usercreatedate IS NOT NULL
  AND usercreatedate != 'NULL'
  AND LEFT(usercreatedate, 10) ~ '^\d{4}-\d{2}-\d{2}$'
GROUP BY creation_date
ORDER BY creation_date;
```

Data Output | Messages | Graph Visualiser | Notifications

| | creation_date date | user_count bigint |
|---|---|---|
| 1 | 2023-01-05 | 3358 |
| 2 | 2023-01-06 | 31 |
| 3 | 2023-01-07 | 18 |
| 4 | 2023-01-08 | 12 |
| 5 | 2023-01-09 | 11 |
| 6 | 2023-01-10 | 17 |
| 7 | 2023-01-11 | 16 |
| 8 | 2023-01-12 | 6 |
| 9 | 2023-01-13 | 10 |
| 10 | 2023-01-14 | 11 |
| 11 | 2023-01-15 | 15 |

Total rows: 783 Query complete 00:00:00.432

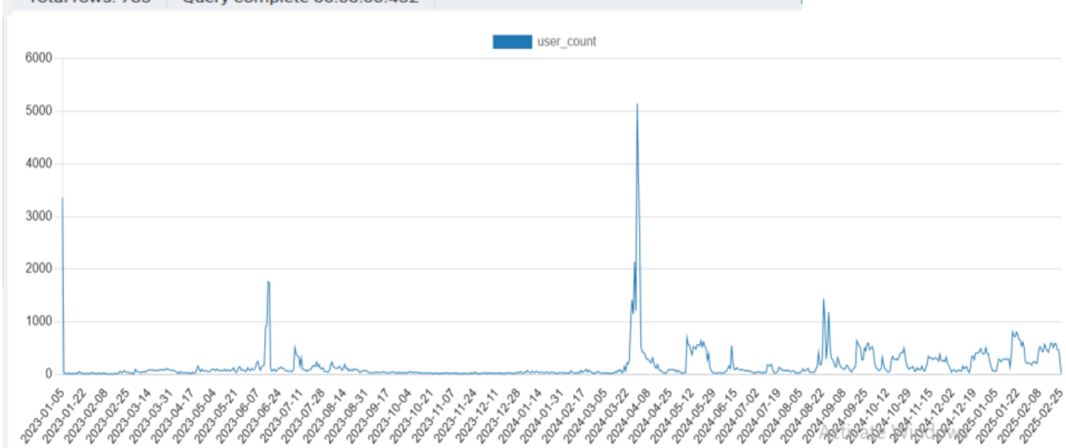
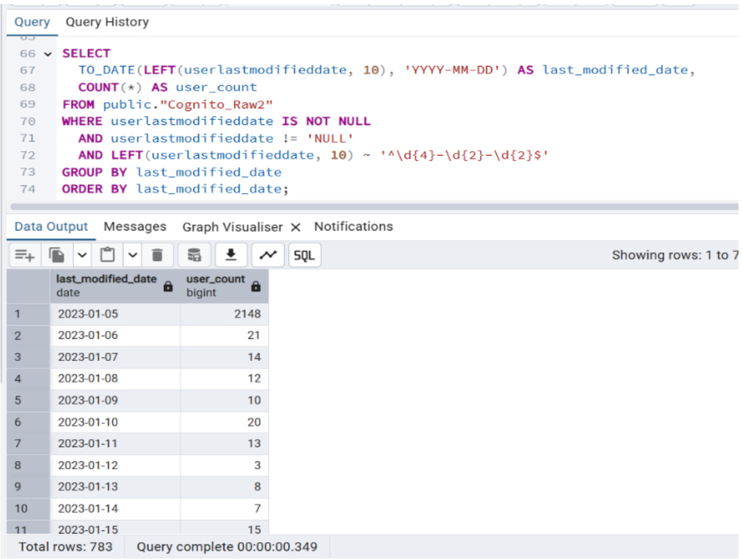



**Figure: Line Chart of "Timeline of Daily User Account Creations" from 'usercreatedate' column.**

**Daily User Account Modifications from 'userlastmodifieddate' column-**

## Overview:



Query | Query History

```
SELECT
    TO_DATE(LEFT(userlastmodifieddate, 10), 'YYYY-MM-DD') AS last_modified_date,
    COUNT(*) AS user_count
FROM public."Cognito_Raw2"
WHERE userlastmodifieddate IS NOT NULL
    AND userlastmodifieddate != 'NULL'
    AND LEFT(userlastmodifieddate, 10) ~ '^\d{4}-\d{2}-\d{2}$'
GROUP BY last_modified_date
ORDER BY last_modified_date;
```

Data Output | Messages | Graph Visualiser | Notifications

Showing rows: 1 to 7

| | last_modified_date date | user_count bigint |
|---|---|---|
| 1 | 2023-01-05 | 2148 |
| 2 | 2023-01-06 | 21 |
| 3 | 2023-01-07 | 14 |
| 4 | 2023-01-08 | 12 |
| 5 | 2023-01-09 | 10 |
| 6 | 2023-01-10 | 20 |
| 7 | 2023-01-11 | 13 |
| 8 | 2023-01-12 | 3 |
| 9 | 2023-01-13 | 8 |
| 10 | 2023-01-14 | 7 |
| 11 | 2023-01-15 | 15 |

Total rows: 783 Query complete 00:00:00.349

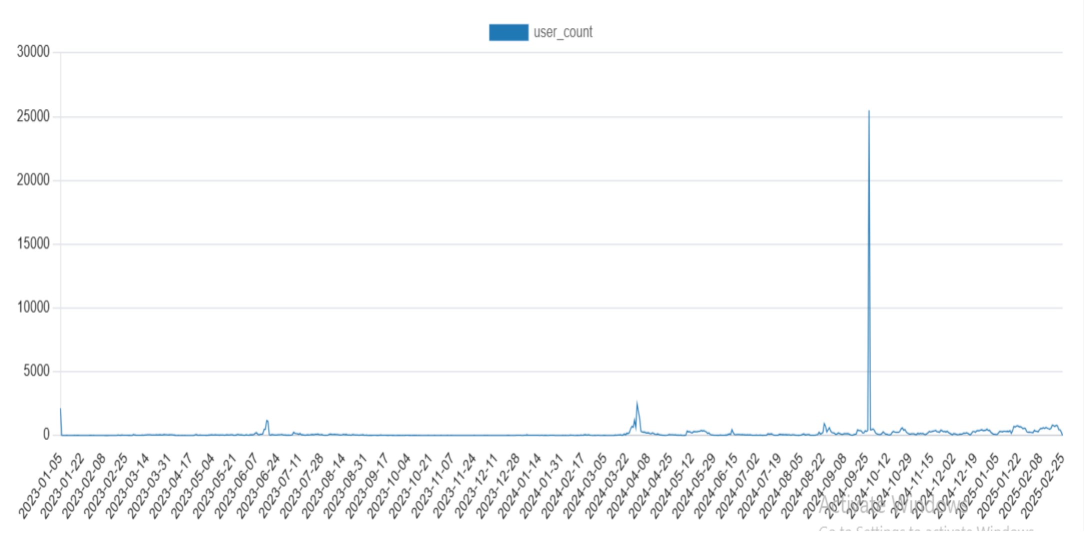



**Figure: Line Chart of "Timeline of Daily User Account Modifications" from 'userlastmodifieddate' column.**

## Exploring Statistic Funtions:

Here's a brief explanation for each statistic functions:

**MIN():** Returns the smallest value in a column.

**MAX():** Returns the largest value in a column.

**MEAN() / AVG():** Calculates the average (mean) value of a numeric column.

**MODE() WITHIN GROUP:** Finds the most frequently occurring value (mode) in a dataset.

**STANDARDDEVIATION():** Computes the sample standard deviation, measuring data spread around the mean.

## Used min(), max() functions in “City” Column:

```sql
SELECT
  MIN(city) AS first_city_alphabetically,
  MAX(city) AS last_city_alphabetically
FROM public."Cognito_Raw2"
WHERE city IS NOT NULL;
```

Data Output Messages Graph Visualiser Notifications

| | first_city_alphabetically text | last_city_alphabetically text |
|---|---|---|
| 1 | a | Zuru |

## Age Visualization by “birthdate column”:

Query Query History

```sql
SELECT
    CASE
        WHEN EXTRACT(YEAR FROM AGE(CURRENT_DATE, TO_DATE(birthdate, 'MM/DD/YYYY'))) < 13 THEN 'Child'
        WHEN EXTRACT(YEAR FROM AGE(CURRENT_DATE, TO_DATE(birthdate, 'MM/DD/YYYY'))) BETWEEN 13 AND 19 THEN 'Teenager'
        WHEN EXTRACT(YEAR FROM AGE(CURRENT_DATE, TO_DATE(birthdate, 'MM/DD/YYYY'))) BETWEEN 20 AND 59 THEN 'Adult'
        ELSE 'Old'
    END AS age_group_label,

    CASE
        WHEN EXTRACT(YEAR FROM AGE(CURRENT_DATE, TO_DATE(birthdate, 'MM/DD/YYYY'))) < 13 THEN '<13'
        WHEN EXTRACT(YEAR FROM AGE(CURRENT_DATE, TO_DATE(birthdate, 'MM/DD/YYYY'))) BETWEEN 13 AND 19 THEN '13-19'
        WHEN EXTRACT(YEAR FROM AGE(CURRENT_DATE, TO_DATE(birthdate, 'MM/DD/YYYY'))) BETWEEN 20 AND 59 THEN '20-59'
        ELSE '60+'
    END AS age_range,

    COUNT(*) AS user_count
FROM public."Cognito_Raw2"
WHERE birthdate ~ '^\d{1,2}/\d{1,2}/\d{4}$'
GROUP BY age_group_label, age_range
ORDER BY age_group_label;
```

Data Output Messages Graph Visualiser Notifications

| | age_group_label text | age_range text | user_count bigint |
|---|---|---|---|
| 1 | Adult | 20-59 | 81001 |
| 2 | Child | <13 | 3 |
| 3 | Old | 60+ | 27 |
| 4 | Teenager | 13-19 | 5285 |

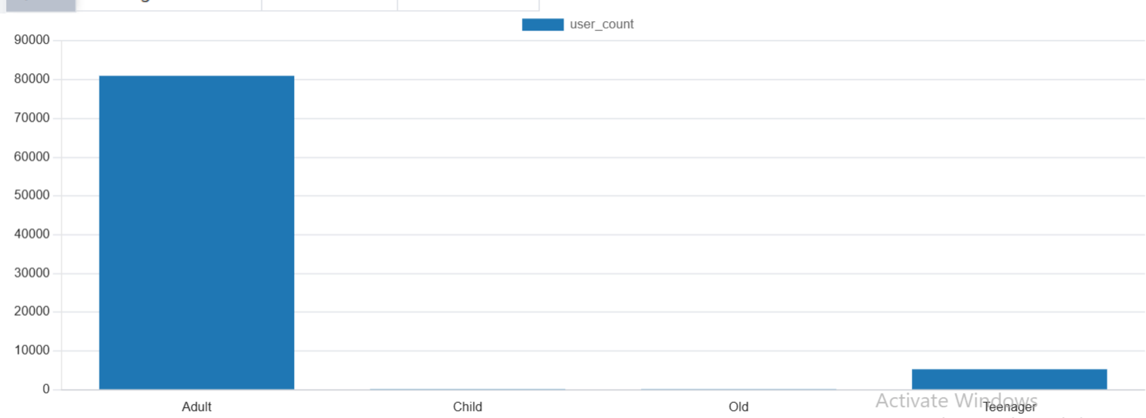



**Figure: Bar Chart of Age Visualization by “birthdate” column.**

## Used mean(), mode() functions in “birthdate” column:

Query Query History

```sql
SELECT
    CASE
        WHEN age < 13 THEN 'Child'
        WHEN age BETWEEN 13 AND 19 THEN 'Teenager'
        WHEN age BETWEEN 20 AND 59 THEN 'Adult'
        ELSE 'Old'
    END AS age_group_label,

    CASE
        WHEN age < 13 THEN '<13'
        WHEN age BETWEEN 13 AND 19 THEN '13-19'
        WHEN age BETWEEN 20 AND 59 THEN '20-59'
        ELSE '60+'
    END AS age_range,

    COUNT(*) AS user_count,
    ROUND(AVG(age), 2) AS mean_age,
    MODE() WITHIN GROUP (ORDER BY age) AS mode_age
FROM (
    SELECT
        EXTRACT(YEAR FROM AGE(CURRENT_DATE, TO_DATE(birthdate, 'MM/DD/YYYY')))::int AS age
    FROM public."Cognito_Raw2"
    WHERE birthdate ~ '^\d{1,2}/\d{1,2}/\d{4}$'
) sub
GROUP BY age_group_label, age_range
ORDER BY age_group_label;
```

Data Output Messages Graph Visualiser Notifications

| | age_group_label text | age_range text | user_count bigint | mean_age numeric | mode_age integer |
|---|---|---|---|---|---|
| 1 | Adult | 20-59 | 81001 | 26.19 | 23 |
| 2 | Child | <13 | 3 | 5.00 | 6 |
| 3 | Old | 60+ | 27 | 69.04 | 60 |
| 4 | Teenager | 13-19 | 5285 | 17.95 | 19 |

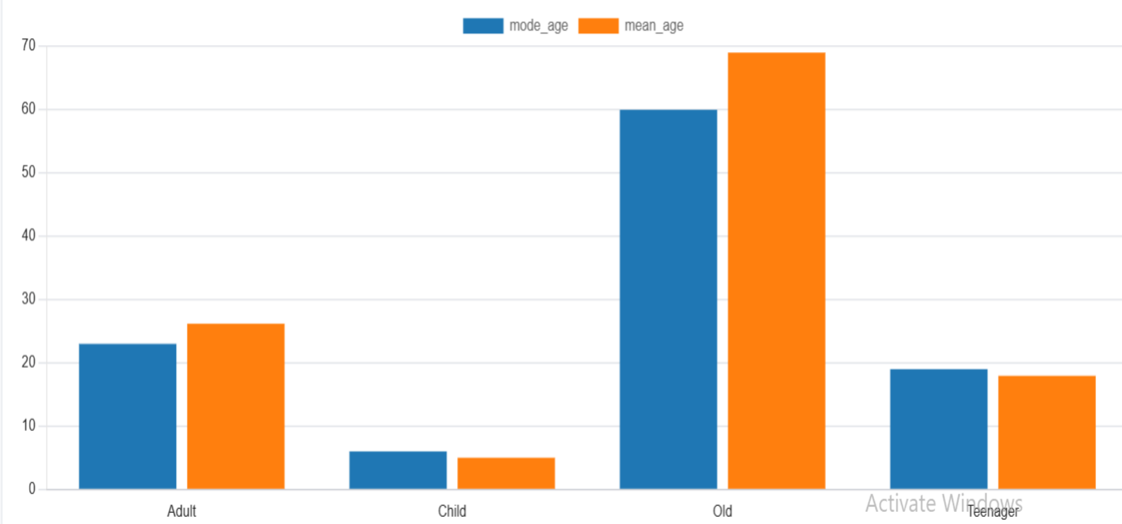



**Figure: Bar Chart of mode_age and mean_age of "birthdate" column.**

## All overview of "birthdate" Column using mean(), mode(), standard deviation() functions:

```sql
WITH age_data AS (
    SELECT
        EXTRACT(YEAR FROM AGE(CURRENT_DATE, TO_DATE(birthdate, 'MM/DD/YYYY'))) AS age
    FROM public."Cognito_Raw2"
    WHERE birthdate ~ '^\d{1,2}/\d{1,2}/\d{4}$'
)
SELECT
    ROUND(AVG(age), 2) AS mean_age,
    MODE() WITHIN GROUP (ORDER BY age) AS mode_age,
    ROUND(STDDEV(age), 2) AS standarddeviation_age
FROM age_data;
```

Data Output Messages Graph Visualiser Notifications

Showing rows: 1 to 1

| | mean_age numeric | mode_age numeric | standarddeviation_age numeric |
|---|---|---|---|
| 1 | 25.70 | 23 | 5.28 |

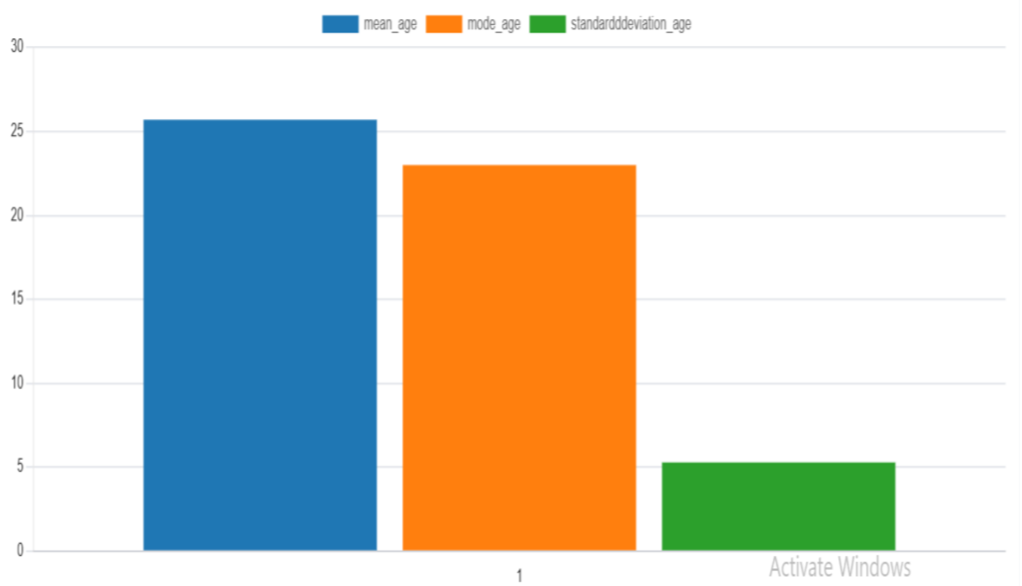



**Figure: Bar Chart of an overview of mode_age , mean_age and standarddeviation_age of everyone's in "birthdate" column.**

## Summary of Findings:

**Key Observations:**

### 1. Incorrect Data Type for Date Columns:

The columns usercreatedate, userlastmodifieddate, and birthdate are currently stored as character varying(200).These columns are meant to represent date-time values but are stored as plain text, which can lead to:

- Inconsistent formats
- Invalid values (e.g., presence of alphabetic strings)
- Difficulties in performing date-based operations such as sorting, filtering, or date arithmetic.

**Recommendation:** Convert these columns to proper TIMESTAMP data types after validating and cleaning the data.

### 2. Invalid Entries in gender Column:

The gender column includes non-standard and potentially harmful values such as:

- URLs (e.g., Don%27t want to specify)
- These suggest missing input validation at the data collection stage.

**Recommendation:** Restrict values to a predefined list (e.g., 'Male', 'Female', 'Other', 'Don't want to specify') and implement input sanitization to prevent injection or data quality issues.

### 3. Non-Numeric Values in zip Column:

The zip column, which should only contain numeric postal codes, has some entries containing alphabetic characters or non-digit symbols.

This indicates inconsistent formatting or entry errors (e.g., ABC123, 7500A, N/A). Difficulties in performing statistic functions.

**Recommendation:** Clean and validate the zip column to ensure it contains only digits. Convert the data type to INTEGER (or VARCHAR with numeric constraints if leading zeros are needed).

4. **Non-Invalid columns:** 'User_id' and 'email' columns has no invalid values. As well as both are containing character varying(200) datatypes correctly with values.

## Next Steps for the "Cognito_Raw2" Dataset:

- ✓ Perform comprehensive data cleaning to remove or correct invalid date, gender, and zip values.
- ✓ Convert usercreatedate, userlastmodifieddate, and birthdate columns from character varying to TIMESTAMP after validation.
- ✓ Standardize the gender column by mapping non-standard entries to a predefined set of valid options.
- ✓ Enforce numeric validation on the zip column and cast it to INTEGER or appropriate text format preserving leading zeros.

Implement validation rules at the data source to prevent future issues related to data types, missing fields, and malformed entries

# Summary of the Report:

The document details Week 1 activities of the **Excelerate Data Visualization Associate Internship**, focusing on **data familiarization**, **PostgreSQL setup**, and **exploratory data analysis (EDA)** across six datasets:

1. **User Data (Learner_Raw.csv):**
   - **Size:** 129,259 rows × 5 columns
   - **Key issues:**
     - Missing values (degree, institution, major > 52k each)
     - 52,692 duplicate rows
     - All columns are text (no numeric fields for advanced stats)
   - **Insights:**
     - Top 10 countries distribution, degree vs. major comparisons
     - Needs data cleaning & standardization before deeper analysis
2. **Opportunity Data (Opportunity_Raw.csv):**
   - **Size:** 188 rows × 5 columns
   - **Key issues:**
     - "NULL" string placeholders in tracking_questions column
     - Duplicate values in non-key columns
   - **Insights:**
     - 8 opportunity categories; variable completeness of tracking data
     - Recommended to replace "NULL" placeholders with true NULL values
     - 

3. **Cohort Data (CohortRaw(in).csv)**

- **Size:** 639 rows × 4 key columns (cohort_code, start_date, end_date, size)
- **Data Quality:**
  - No missing values or duplicates

- Dates converted from Unix timestamps to standard datetime

- **Statistics:**
  - **Cohort size:** Mean = 5,741, Median = 800, Max = 100,000 (heavily right-skewed with few very large cohorts)
  - **Duration:** Mean = 56 days, Median = 29 days, Max = 1,096 days (high variability, most short-lived)
- **Insights:**
  - Growth is event-driven (sharp spikes in late 2022 & Aug 2023) rather than organic
  - Few long-duration cohorts suggest valuable, loyal user groups
  - Majority of cohorts are small and short-lived, likely tied to specific campaigns

4. **Marketing Data (Marketing_Raw.csv):**
   - **Size:** 141 rows × 13 columns
   - **Key issues:**
     - Blank values in campaign_name, outbound_clicks, and outbound_type
     - Logical inconsistency: reach = 0 while clicks > 0
   - **Insights:**
     - Strong correlation between reach & results, and between budget & clicks
     - Single-day snapshot (2023-01-01), so no time trend possible
5. **Learner Opportunity Data (learner_opportunity_raw.csv):**
   - **Size:** 113,602 rows × 5 columns
   - **Key issues:**
     - Missing values: assigned_cohort (13,318), apply_date (188), status (186)
     - Uses numeric codes for status & assigned_cohort (needs reference mapping)
   - **Insights:**
     - Monthly enrollment trends show growth with seasonal peaks (Jan 2025 highest)
6. **Cognito Data (Cognito_Raw2.xlsx):**
   - **Size:** 129,179 rows × 9 columns
   - **Key issues:**
     - Missing values (gender, birthdate, city, zip, state ~ 42k each)
     - Incorrect data types for date fields (stored as text)
     - Invalid entries in gender and zip columns
   - **Insights:**
     - Top cities/states identified, account creation/modification timelines plotted